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ESTIMATION OF NONLINEAR MODELS WITH MEASUREMENT ERROR

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This paper presents a solution to an important econometric problem, namely the root n consistent estimation of nonlinear models with measurement errors in the explanatory variables, when one repeated observation of each mismeasured regressor is available. While a root n consistent estimator has been derived for polynomial specifications (see Hausman, Ichimura, Newey, and Powell (1991)), such an estimator for general nonlinear specifications has so far not been available. Using the additional information provided by the repeated observation, the suggested estimator separates the measurement error from the "true" value of the regressors thanks to a useful property of the Fourier transform: The Fourier transform converts the integral equations that relate the distribution of the unobserved "true" variables to the observed variables measured with error into algebraic equations. The solution to these equations yields enough information to identify arbitrary moments of the "true," unobserved variables. The value of these moments can then be used to construct any estimator that can be written in terms of moments, including traditional linear and nonlinear least squares estimators, or general extremum estimators. The proposed estimator is shown to admit a representation in terms of an influence function, thus establishing its root n consistency and asymptotic normality. Monte Carlo evidence and an application to Engel curve estimation illustrate the usefulness of this new approach.

KEYWORDS: Measurement error, errors-in-variables, Fourier transform, nonlinear models.

1. INTRODUCTION

THE ASSUMPTION THAT THE REGRESSORS are measured without error is often made on the basis of convenience, rather than being based on a formal justification. However, due in part to the increased availability of microeconomic datasets, the problem of measurement errors is receiving increased attention. There is also a clear trend, among both theoretical and applied econometricians, toward the use of nonlinear models. Unfortunately, a difficult problem arises at the intersection of these two trends. A nonlinear function of a mismeasured variable introduces an error term that is not additively separable from the true value of the regressor, making standard linear instrumental variable (IV) estimators inconsistent. As a result, in general nonlinear specifications, there exists so far no estimator that exhibits all the desirable properties of estimators that correct for measurement error in linear specifications: root n consistency, asymptotic normality, absence of distributional assumptions, and weak regularity conditions. The present paper intends to fill this gap.

In order to better perceive the complexity of this task, it is instructive to overview the numerous partial solutions to the measurement error problem

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in nonlinear specifications that have been devised in recent years.[2] Very effective ways to handle the errors-in-variables problem in polynomial specifications have been developed by Hausman, Ichimura, Newey, and Powell (1991), both in the presence of instruments and repeated observations. Other functional forms have been considered by Lewbel (1996), who presents a way to correct for the presence of measurement error for a class of Engel curve specifications through the use of instruments, but the derivation of the estimator's asymptotic properties relies on distributional assumptions.

Useful results have been obtained for general nonlinear specifications as well. Newey (2001) devises a consistent IV estimator based on the method of simulated moments. Unfortunately, unless distributional assumptions regarding the measurement error are made, its asymptotic properties are not fully characterized. A related estimator due to Wang and Hsiao (1995) alleviates some of these problems, but at the expense of the restrictive assumption of absolute integrability of the regression function.

The case when one repeated measurement is available has also been studied. Hausman, Newey, and Powell (1995) introduce a two-step estimator, based on their polynomials results (Hausman, Ichimura, Newey, and Powell (1991)), which enables the estimation of arbitrary nonlinear specifications. Unfortunately, strong restrictions on the existence of various moments are needed and, aside from consistency, its asymptotic properties are not characterized. Li (2002) addresses the same problem through a different route, using an identity due to Kotlarski (see Rao (1992, p. 21)) to obtain an estimate of the density of the measurement error that can then be used to remove the measurement error bias, thanks to a technique devised by Hsiao (1989). However, a relatively large number of statistical independence assumptions are needed and the asymptotic distribution of the estimator is not characterized.

The measurement error problem in estimators other than least squares has received relatively little attention. Useful results have been derived in Lewbel (1998) for generalized linear models and in Weiss (1993) for a censored linear regression model. Estimates of the measurement error bias in nonlinear models, in the limiting case where the measurement error variance is small, are

[2]Although the use of instruments or repeated measurements is, by far, the most common way to overcome measurement error problems, alternative approaches have been suggested. In polynomial specifications, the use of information provided by higher order moments of the observed variables has been proposed by Chesher (1998) and Lewbel (1997). The use of additional information regarding the magnitude and the distribution of measurement errors is another possibility (see, for instance, Cheng and Schneeweiss (1998) for polynomial least squares models, or Wang (1998) for censored linear errors-in-variables models). The availability of validation data also leads to identification (see, for instance, Lee and Sepanski (1995)). A number of authors have used the "shrinking covariance approximation" to construct consistent estimators in the presence of measurement errors (see, for instance, Amemiya (1985)). Carroll, Ruppert, and Stefanski (1995) present an overview of various approaches with a particular focus on approximately consistent methods.

also available (Chesher (1991), Carroll, Ruppert, and Stefanski (1995)). But so far, there exists no general framework to convert an arbitrary estimator that assumes that all regressors are perfectly observed into one that is robust to the presence of measurement errors in the regressors.

This paper presents a general solution to the errors in variables problem in general nonlinear models when one repeated observation is available for each mismeasured variable. We introduce a well defined procedure to convert an existing estimator that is root n consistent and asymptotically normal in the absence of measurement errors into an estimator that is also root n consistent and asymptotically normal in the presence of measurement error. The proposed solution overcomes most limitations of existing approaches, as it requires mild regularity conditions, no distributional assumptions, and few independence assumptions. Our approach relies on the fact that a large class of models can be identified from the knowledge of a set of sample moments. While the required moments may involve unobserved variables, we show that the repeated measurement provides enough information to identify them. The proposed estimator takes the form of a functional of the empirical characteristic functions of the observed variables. The influence function approach is used to show root n consistency and asymptotic normality of the estimator.

This paper is organized as follows. To avoid notational complications that would obscure the main features of the proposed estimator, we first present a simple treatment of the measurement error problem in least squares regressions that include a nonlinear function of one mismeasured regressor and any number of perfectly observed regressors. We then derive the general theory for the treatment of the measurement error problem in an extremely wide class of estimators, including nonlinear least squares estimators, GMM estimators, and most maximum likelihood estimators with an arbitrary number of mismeasured regressors. We finally show the practical usefulness of the approach introduced in this paper through both Monte Carlo experiments and an application to Engel curve estimation.

2. LEAST SQUARES ESTIMATORS

2.1. *The Model*

We start by considering a simple nonlinear errors-in-variable model of the form

$$
\begin{aligned}
&y = \sum_{k=1}^{M} \beta_k h_k(x^*) + \sum_{l=1}^{L} \beta_{l+M} w_l + \Delta y,\\
(1)\qquad &x = x^* + \Delta x,\\
&z = x^* + \Delta z,
\end{aligned}
$$

where β is the parameter vector to be estimated and the scalar variables y, x, z, and w_l for $l = 1, \dots, L$ are observed while the scalar variables x^*, Δx, Δy, and Δz are not. The variables x and z are error-contaminated measurements of the true regressor x^*, while Δx, Δy, and Δz are disturbances. For the benefit of conciseness, let us adopt the convention that $w_0 \equiv y$, so that all the variables that do not require a correction for measurement error can be referred to as w_l for $l = 0, \dots, L$. Aside from weak regularity conditions to be described later, the shape of the joint density of x, z, Δx, Δy, w is left unrestricted. These variables are assumed to satisfy the following assumptions, which are similar to the ones considered by Hausman, Ichimura, Newey, and Powell (1991).

ASSUMPTION 1:

$$
\begin{aligned}
&E[\Delta y | x^*, \Delta z] = 0,\\
&E[\Delta x | x^*, \Delta z] = 0,\\
&\Delta z \textit{ and } x^* \textit{ are mutually independent},\\
&E[w_l | x^*, \Delta z] = E[w_l | x^*] \quad \textit{for } l = 1, \dots, L.
\end{aligned}
\tag{2}
$$

While the disturbances on the two first equations of Model (1) are only required to satisfy mild conditional mean assumptions, the assumption of independence between the true regressor x^* and the disturbance Δz is relatively strong. However, this independence is required by the nonlinearity of the model—all existing estimators of nonlinear models with measurement error require at least one independence assumption, and usually more than one. It is particularly useful for practical applications that independence between the two measurement errors Δx and Δz is not required, since it is easy to imagine situations where their variances would be related. The measurement errors Δx and Δz are, however, required to be *mean independent* ($E[\Delta x | \Delta z] = 0$), as implied by the assumption that $E[\Delta x | x^*, \Delta z] = 0$.

There is some asymmetry between the assumptions on the two measurements, but that is done so that the results can be derived under the weakest possible assumptions. In some applications, it may be possible to impose the independence assumptions on both measurements of x^* and adjust the remaining conditional mean assumptions, thus obtaining an overidentified model. The assumptions of the model can then be tested by comparing the two estimates obtained by exchanging the role of x and z, or a more efficient estimate of β can be obtained by forming a weighted average of the two resulting estimates. When more than two measurements are available, they can, of course, be combined pairwise and the estimates obtained from each pairwise combination can then be averaged to yield a more accurate estimate than would be possible with only two measurements. Only one of those repeated measurements needs to have an independent measurement error, although if more than one has, a larger number of pairwise combinations are possible. The overidentification provided by the availability of more than two measurements also enables testing of the model's assumptions.

While less prevalent than instruments, repeated measurements have found a number of applications in the empirical econometric literature (Ashenfelter and Krueger (1994), Hausman, Newey, and Powell (1995), Morey and Waldman (1998), Bowles (1972), Borus and Nestel (1973), Freeman (1984)[3]) and in sociology (as noted by Griliches (1986)). Although one can always question whether two observations are truly repeated measures of the same quantity, such issues are similar to the more familiar concerns regarding the validity of instruments. In fact, our assumptions tolerate various imperfections in the repeated measurement z. Because the density of the two disturbances Δx and Δz can differ, the repeated observation z could be an imperfect measurement of an "updated" value of the regressor x^*. The error on the repeated measurement would then take the form $\Delta z = \epsilon + \nu$, where ϵ is an "innovation" that is independent from x^*, while ν is a genuine measurement error, also assumed independent from x^*. In addition, since the error term Δz does not need to have zero mean, our assumptions allow for a systematic drift between the two measurements. In short, even a biased measurement of an imperfect replicate observation can be used as a repeated measurement within our framework. For instance, in the context of time series, whenever one has access to a time series generated by a random walk process shifted by a deterministic trend, two consecutive mismeasured observations of the series can be used as a pair of measurements of the true value of the first observation. These observations considerably expand the set of valid repeated measures.

2.2. *Derivation of the Estimator*

An important feature of our estimation procedure is that we clearly separate the measurement error problem from the conventional least square minimization procedure. Since the probability limit of the objective function associated with Model (1) is $E[(y - \sum_{k=1}^{M} \beta_k h_k(x^*) - \sum_{l=1}^{L} \beta_{l+M} w_l)^2]$, the parameter vector β can be identified from the knowledge of moments of the form

$$E[w_l w_{l'}] \quad \text{for } l, l' = 0, 1, \ldots, L, \tag{3}$$

which can directly be estimated from sample averages, and moments of the form

$$\begin{aligned} &E[h_k(x^*) h_{k'}(x^*)] \quad \text{for } k, k' = 1, \ldots, M \quad \text{and} \\ &E[w_l h_k(x^*)] \quad \text{for } l = 0, \ldots, L \text{ and } k = 1, \ldots, M, \end{aligned} \tag{4}$$

[3]Freeman's dataset (the January 1977 Employer Employee Matched Sample, Current Population Survey) contains wages reported by employers and employees, which are perfect examples of repeated measurements.

which involve the unobserved variable x^*. All the moments in equation (4) have the general form $E[Wu(x^*)]$ where $u(x^*)$ is one of $\{h_k(x^*) \times h_{k'}(x^*)\}_{k,k'=1,\ldots,M}$ when $W = 1$, or $u(x^*)$ is one of $\{h_k(x^*)\}_{k=1,\ldots,M}$ when $W = w_0, \ldots, w_L$. Our task thus consists of identifying moments of the form $E[Wu(x^*)]$ based on the knowledge of the observed value x and a repeated observation z. We also take the convention that integrals without integration bounds extend from $-\infty$ to $+\infty$. Some basic regularity conditions need to be imposed before our fundamental result for identification can be given.[4]

ASSUMPTION 2: *$E[|x^*|]$, $E[|\Delta x|]$, and $E[|w_l|]$ for $l = 0, \ldots, L$ are finite.*

THEOREM 1: *Given Model* (1) *and under Assumptions* 1 *and* 2, *for $W = 1$, $w_0, \ldots, w_L$ and any function $u : \mathbb{R} \mapsto \mathbb{R}$ such that $E[Wu(x^*)]$ exists,*

$$E[Wu(x^*)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mu_u(-\chi)\phi_W(\chi)\,d\chi \equiv \frac{1}{2\pi} \int \mu_u(-\chi)\phi_W(\chi)\,d\chi, \tag{5}$$

where $\mu_u(\chi)$ is the Fourier transform of $u(x^*)$ and

$$\phi_W(\chi) = \frac{E[We^{i\chi z}]}{E[e^{i\chi z}]} \exp\left(\int_0^{\chi} i\frac{E[xe^{i\zeta z}]}{E[e^{i\zeta z}]}\,d\zeta\right), \tag{6}$$

where $i = \sqrt{-1}$. Moreover, $\phi_W(\chi) = E[We^{i\chi x^*}]$. In particular, $\phi_1(\chi)$ is the characteristic function of x^*.

PROOF: In order to identify moments of the form $E[u(x^*)]$ and $E[w_l u(x^*)]$ we need to express, for any l in $\{0, \ldots, L\}$, the joint density of w_l and x^*, denoted $s(w_l, x^*)$, in terms of the known density of w_l, x, and z, denoted $p(w_l, x, z)$. To achieve this, we first relate these two densities to $f(w_l, x^*, \Delta x, \Delta z)$, the more fundamental density that embodies the information regarding all the variables in the model. On the one hand,

$$s(w_l, x^*) = \iint f(w_l, x^*, \Delta\tilde{x}, \Delta\tilde{z})\,d(\Delta\tilde{x})\,d(\Delta\tilde{z}), \tag{7}$$

which can be compactly expressed in terms of log characteristic functions (see, for instance, Loève (1977), Arsac (1966)):

$$\Sigma(\omega, \chi) = \Phi(\omega, \chi, 0, 0), \tag{8}$$

[4]Equation (6) is a generalization of an identity derived by Kotlarski (see Rao (1992, p. 21)). Our proof of this result requires much weaker independence assumptions. In particular, we do not require independence between Δx and x^* and, more importantly, between Δx and Δz.

where Σ and Φ are the natural logarithms of the Fourier transforms of s and f, respectively. On the other hand, the relationship between p and f is

$$p(w_l, x, z) = \int f(w_l, \tilde{x}^*, x - \tilde{x}^*, z - \tilde{x}^*) d\tilde{x}^*. \tag{9}$$

Once again, this identity is naturally expressed in terms of Fourier transforms:

$$\begin{aligned}
\pi(\omega, \xi, \zeta) &= \iiiint f(\tilde{w}_l, \tilde{x}^*, \tilde{x} - \tilde{x}^*, \tilde{z} - \tilde{x}^*) e^{i(\xi\tilde{x} + \zeta\tilde{z} + \omega\tilde{w}_l)} d\tilde{w}_l d\tilde{z} d\tilde{x} d\tilde{x}^* \\
&= \iiint \phi^{[234]}(\omega, \tilde{x}^*, \tilde{x} - \tilde{x}^*, \tilde{z} - \tilde{x}^*) e^{i(\xi\tilde{x} + \zeta\tilde{z})} d\tilde{z} d\tilde{x} d\tilde{x}^* \\
&= \iint \phi^{[23]}(\omega, \tilde{x}^*, \tilde{x} - \tilde{x}^*, \zeta) e^{i\zeta\tilde{x}^*} e^{i\xi\tilde{x}} d\tilde{x} d\tilde{x}^* \\
&= \int \phi^{[2]}(\omega, \tilde{x}^*, \xi, \zeta) e^{i\zeta\tilde{x}^*} e^{i\xi\tilde{x}^*} d\tilde{x}^* \\
&= \phi(\omega, \xi + \zeta, \xi, \zeta)
\end{aligned} \tag{10}$$

where π and ϕ are the Fourier transforms of p and f, respectively. Bracketed superscripts on ϕ denote the arguments with respect to which the Fourier transform has not yet been applied. No particular assumptions are required to ensure the existence of these Fourier transforms, because probability densities are absolutely integrable and their Fourier transforms therefore exist. Taking the logarithm on each side preserves the equality:

$$\Pi(\omega, \xi, \zeta) = \Phi(\omega, \xi + \zeta, \xi, \zeta), \tag{11}$$

where Π and Φ are the natural logarithms of π and ϕ, respectively.[5]

Equipped with equations (8) and (11), we can now determine moments of the form $E[u(x^*)]$ in terms of $\Pi(\omega, \xi, \zeta)$. Since the density of w_l is not needed for this exercise, our task is to find $\Phi(0, \chi, 0, 0)$ in terms of $\Pi(\omega, \xi, \zeta)$. We first note that differentiating equation (11) with respect to ξ yields

$$\Pi^{(2)}(\omega, \xi, \zeta) = \Phi^{(2)}(\omega, \xi + \zeta, \xi, \zeta) + \Phi^{(3)}(\omega, \xi + \zeta, \xi, \zeta), \tag{12}$$

$$\Pi^{(2)}(0, 0, \zeta) = \Phi^{(2)}(0, \zeta, 0, \zeta) + \Phi^{(3)}(0, \zeta, 0, \zeta), \tag{13}$$

where a numeral superscript k in parentheses denotes a derivative with respect to the kth argument. The required derivatives exist, by Assumption 2. Since, by Assumption 1, $E[\Delta x | x^*, \Delta z] = 0$, we have that $\Phi^{(3)}(0, \chi, 0, \zeta) = 0$ and the

[5] We select one branch of the logarithm function such that $\Phi(0, 0, 0, 0) = 0$ and $\Pi(0, 0, 0) = 0$.

second term of the right-hand side vanishes. The independence of x^* and Δz (from Assumption 1) lets us write

$$\Phi(0,\chi,0,\zeta)=\Phi(0,\chi,0,0)+\Phi(0,0,0,\zeta). \tag{14}$$

Differentiating this expression with respect to χ *before* setting $\zeta=\chi$ yields

$$\Phi^{(2)}(0,\chi,0,\chi)=\Phi^{(2)}(0,\chi,0,0). \tag{15}$$

Equation 13 then becomes

$$\Pi^{(2)}(0,0,\zeta)=\Phi^{(2)}(0,\zeta,0,0). \tag{16}$$

The log characteristic function $\Phi(0,\chi,0,0)$ can then be determined by integrating equation (16) and noting that $\Phi(0,0,0,0)=0$:

$$\Phi(0,\chi,0,0)=\int_0^{\chi}\Phi^{(2)}(0,\zeta,0,0)\,d\zeta, \tag{17}$$

where

$$\Phi^{(2)}(0,\zeta,0,0)=\Pi^{(2)}(0,0,\zeta)=\frac{\pi^{(2)}(0,0,\zeta)}{\pi(0,0,\zeta)}=\frac{iE[xe^{i\zeta z}]}{E[e^{i\zeta z}]}. \tag{18}$$

The last equality follows from a simple generalization of the Moment Theorem ($d^kE[e^{i\zeta z}]/d\zeta^k=E[(iz)^k e^{i\zeta z}]$).

Noting that $\exp(\Phi(0,\chi,0,0))=\phi(0,\chi,0,0)$, the characteristic function of x^*, we can express $E[u(x^*)]$ entirely in Fourier representation with the help of Parseval's Identity:

$$\begin{aligned}E[u(x^*)]&=\int u(\tilde{x}^*)\left(\iiint f(\tilde{w}_l,\tilde{x}^*,\Delta\tilde{x},\Delta\tilde{z})\,d\tilde{w}_l\,d(\Delta\tilde{x})\,d(\Delta\tilde{z})\right)d\tilde{x}^*\\&=\frac{1}{2\pi}\int\mu_u(-\chi)\phi(0,\chi,0,0)\,d\chi=\frac{1}{2\pi}\int\mu_u(-\chi)\phi_1(\chi)\,d\chi,\end{aligned} \tag{19}$$

where $\mu_u(\chi)$ is the Fourier transform of $u(x^*)$ and $\phi_1(\chi)\equiv\phi(0,\chi,0,0)$. In practical examples, $\mu_u(\chi)$ may be a distribution[6] (also called a generalized function) instead of a function in the usual sense. This does not pose any fundamental problem, as long as the inner product between $\mu_u(\chi)$ and $\phi(0,\chi,0,0)$ remains finite, a requirement explicitly imposed by the hypothesis that $E[Wu(x^*)]$ exists.

[6]In this paragraph, the word "distribution" is not used in the usual probabilistic sense, but rather in the sense of the theory of distributions, namely, it is an element of a set of bounded linear functionals.

We now focus on the other type of moments needed for identification, that is, $E[w_l u(x^*)]$ for some $l \in \{0, \ldots, L\}$. Since only an expectation of w_l, and not its full distribution, is needed, our task is simplified: A simple application of the Moment Theorem shows that we do not need to know $\Phi(\omega, \zeta, 0, 0)$ for all ω, but only $\Phi^{(1)}(0, \zeta, 0, 0)$. Our starting point is again a differentiation of equation (11), but now with respect to ω:

$$\Pi^{(1)}(\omega, \xi, \zeta) = \Phi^{(1)}(\omega, \xi + \zeta, \xi, \zeta), \tag{20}$$

$$\Pi^{(1)}(0, 0, \zeta) = \Phi^{(1)}(0, \zeta, 0, \zeta). \tag{21}$$

The required derivatives exist, by Assumption 2.

The remainder of the proof relies on $E[w_l|x^*, \Delta z] = E[w_l|x^*]$. While Assumption 1 already implies that this is the case for $l = 1, \ldots, L$, we need to show that this equality holds for $w_0 = y$ as well.

Let $y^* = \sum_{k=1}^{M} \beta_k h_k(x^*) + \sum_{l=1}^{L} \beta_{l+M} w_l$. From Assumption 1, $E[\Delta y|x^*, \Delta z] = 0$, and thus, $E[\Delta y|x^*] = E[E[\Delta y|x^*, \Delta z]|x^*] = 0$. It follows that $E[y|x^*, \Delta z] = E[y^*|x^*, \Delta z] + E[\Delta y|x^*, \Delta z] = E[y^*|x^*] + 0 = E[y^*|x^*] + E[\Delta y|x^*] = E[y|x^*]$. This equality between $E[w_l|x^*, \Delta z]$ and $E[w_l|x^*]$ can be expressed in Fourier representation:

$$\Phi^{(1)}(0, \chi, 0, \zeta) = \Phi^{(1)}(0, \chi, 0, 0) \quad \forall \chi, \zeta \tag{22}$$

and equation (21) becomes

$$\Pi^{(1)}(0, 0, \chi) = \Phi^{(1)}(0, \chi, 0, 0). \tag{23}$$

With the help of Parseval's Identity and the Moment Theorem, the desired expectation can then be written:

$$\begin{aligned} E[w_l u(x^*)] &= \int u(\tilde{x}^*)\left(\iiint \tilde{w}_l f(\tilde{w}_l, \tilde{x}^*, \Delta\tilde{x}, \Delta\tilde{z})\, d\tilde{w}_l\, d(\Delta\tilde{x})\, d(\Delta\tilde{z})\right) d\tilde{x}^* \\ &= \frac{-i}{2\pi}\int \mu_u(-\chi)\phi^{(1)}(0, \chi, 0, 0)\, d\chi \\ &= \frac{-i}{2\pi}\int \mu_u(-\chi)\phi(0, \chi, 0, 0)\Phi^{(1)}(0, \chi, 0, 0)\, d\chi \\ &= \frac{-i}{2\pi}\int \mu_u(-\chi)\phi(0, \chi, 0, 0)\Pi^{(1)}(0, 0, \chi)\, d\chi, \end{aligned} \tag{24}$$

where

$$\Pi^{(1)}(0, 0, \chi) = \frac{\pi^{(1)}(0, 0, \chi)}{\pi(0, 0, \chi)} = \frac{iE[w_l e^{i\chi z}]}{E[e^{i\chi z}]} \tag{25}$$

and $\phi(0,\chi,0,0)=\exp(\Phi(0,\chi,0,0))$, where $\Phi(0,\chi,0,0)$ is given, as before, by equations (17) and (18).

Collecting the results of equations (17), (18), (24), and (25) and setting $\phi_W(\chi)\equiv\phi(0,\chi,0,0)E[We^{i\chi z}]/E[e^{i\chi z}]$ for $W=1,w_0,\dots,w_L$ provides the conclusion of the theorem. *Q.E.D.*

We note that the right-hand sides of equations (5) and (6) only involve expectations of observable variables that are clearly identifiable. For future reference, we will denote these identifiable moments by

$$m_a(\zeta)=E[ae^{i\zeta z}] \tag{26}$$

for $a=1,x,w_0,\dots,w_L$, while the sample analogue of these moments will be denoted by $\hat{m}_a(\zeta)$. We will also denote by "hats" any quantities derived from these sample moments using equations (5) and (6).

2.3. *Polynomial Specifications*

It is interesting to show that the identification result stated in Theorem 1 is a generalization of the treatment of polynomial functional forms suggested by Hausman, Ichimura, Newey, and Powell (1991). When $u(x^*)=(x^*)^s$ for $s\in\mathbb{N}$, its Fourier transform, $\mu_u(\chi)$, is a distribution $\delta^{(s)}(\chi)$, such that

$$E[(x^*)^s]=\frac{1}{2\pi}\int\delta^{(s)}(-\chi)\phi_1(\chi)d\chi=(-i)^s\left.\frac{d^s\phi_1(\chi)}{d\chi^s}\right|_{\chi=0}, \tag{27}$$

where $\phi_1(\chi)$ is the characteristic function of x^*. It can be shown that substituting in the expression of $\phi_1(\chi)$ given by equation (6) and evaluating the derivative present in equation (27) through a recursive procedure yields an estimator of $E[(x^*)^s]$ that is identical to the one suggested in Hausman, Ichimura, Newey, and Powell (1991). A similar treatment provides the values of $E[(x^*)^s w_l]$. Note that, in the particular case of linear specifications, this estimator reduces to the well-known linear instrumental variable (IV) estimator, where the repeated measurement is used as an instrument.[7]

This relationship with polynomial errors-in-variable regressions helps explain the difficulty of handling general functional forms through polynomial approximations, as suggested in Hausman, Newey, and Powell (1995). Such an approach relies on the assumption that the characteristic function of the unobserved regressor can be extrapolated over all frequencies from its behavior at zero frequency, a requirement that demands strong regularity conditions. Of course, an absence of asymptotic results, aside from consistency, further limits the usefulness of the polynomial approach.

[7]Using a repeated measurement as an instrument is only possible in linear specifications.

2.4. *Asymptotic Properties*

We present the asymptotic properties of this estimator in two parts. We first provide a very general asymptotic normality and root n consistency result for estimated moments that is very useful to establish the asymptotics of a wide class of estimators. We then specialize this result to handle the simple case of the least square estimator we have considered so far.

We need the following assumptions.

ASSUMPTION 3: $(x_j, z_j, w_{0j}, \ldots, w_{Lj})'$ *for* $j = 1, \ldots, n$ *is an iid sample of the random variables* $(x, z, w_0, \ldots, w_L)'$.

ASSUMPTION 4: $E[|x|^{2-k}|z|^k] < \infty$ *and* $E[|w_l|^{2-k}|z|^k] < \infty$ *for* $l = 0, \ldots, L$ *and* $k = 0, 1$.

ASSUMPTION 5: $|m_1(\zeta)| > 0$ *for finite* ζ.

Assumption 3 is convenient, as it enables the use of the Lindeberg–Lévy Central Limit Theorem. It can, however, be relaxed through the use of more general Central Limit Theorems. Assumption 4 imposes very mild conditions regarding the existence of moments of the observable variables. The requirement that the characteristic function of z be nonvanishing, imposed in Assumption 5, is very common in the deconvolution literature (Carroll, Ruppert, and Stefanski (1995), Fan (1991), Fan and Truong (1993), Li and Vuong (1998), Li (2002), Horowitz and Markatou (1996)).

The following definitions are helpful in order to specify the regularity conditions needed.

DEFINITION 1: For a given $s \in \mathbb{N}$ and a given $\bar{\zeta} \in \mathbb{R}^+$, let $\mathcal{D}_{s,\bar{\zeta}}$ be the set of all (generalized) functions $\mu : \mathbb{R} \mapsto \mathbb{R}$ such that, for some given $A' \in \mathbb{R}^+$,

$$(28) \qquad \int_{-\bar{\zeta}}^{\bar{\zeta}} \left| \int_{-\bar{\zeta}}^{\chi_s} \cdots \int_{-\bar{\zeta}}^{\chi_1} \mu(\chi_0) d\chi_0 \ldots d\chi_{s-1} \right| d\chi_s \le A'.$$

DEFINITION 2: Let $\mathbb{S}$ be the set of all functions $u : \mathbb{R} \mapsto \mathbb{R}$ whose Fourier transform $\mu_u(\chi)$ is such that: (i) for some given $\bar{\zeta}, A \in \mathbb{R}^+$ and $\eta \in]0, 1[$,

$$(29) \qquad \int_{\zeta}^{\infty} |\mu_u(\chi)| d\chi \le A(\lambda(\zeta))^{-1-\eta},$$

for all $\zeta \ge \bar{\zeta}$, where, for $m_1(\cdot), m_x(\cdot)$ as defined in equation (26),

$$(30) \qquad \lambda(\zeta) = (1 + |\zeta|) \left(\sup_{\chi \in [-\zeta, \zeta]} \frac{|m_x(\chi)|}{|m_1(\chi)|} \right) \left(\sup_{\chi \in [-\zeta, \zeta]} |m_1(\chi)|^{-1} \right),$$

and (ii) there exists $s \in \mathbb{N}$ such that $\mu_u \in \mathcal{D}_{s,\bar{\zeta}}$, and $E[|a|^2|z|^{2s}]$ exists for $a = 1, x, w_0, \ldots, w_L$.

Definition 1 defines a set of generalized functions that may include singularities, such as derivatives of delta functions $\delta^{(s)}(\chi)$ (as in equation (27)), and is therefore sufficiently rich to include the Fourier transform of most functions $u(x^*)$ one is likely to encounter in practice. Definition 2 constrains the tail behavior of $\mu_u(\chi)$, which amounts to imposing smoothness restrictions on $u(x^*)$, as will be further discussed below.

We will now establish the asymptotic properties of estimated moments $\tilde{E}[Wu(x^*)]$ defined as follows.

DEFINITION 3: For $W = 1, w_0, \ldots, w_L$, let $\hat{\phi}_W(\chi)$ be obtained by substituting sample moments into equation (6), let $\hat{\alpha}_W = n^{-1}\sum_{j=1}^n |W_j|$, where $W_1, \ldots, W_n$ denotes an iid sample of the random variable W, and let

$$\tilde{\phi}_W(\chi) = \hat{\phi}_W(\chi)\min\{1, 2\hat{\alpha}_W/|\hat{\phi}_W(\chi)|\}. \tag{31}$$

Then $\tilde{E}[Wu(x^*)]$ is given by equation (5) with $\phi_W(\chi)$ replaced by $\tilde{\phi}_W(\chi)$.

The constraint imposed by equation (31) is based on the empirical analogue of the trivially satisfied inequality $|\phi_W(\chi)| = |E[We^{i\chi x^*}]| \leq AE[|W|]$ for any $A \geq 1$. This constraint is introduced to ensure that the estimated $\tilde{\phi}_W(\chi)$ is uniformly bounded in probability, even though $\hat{\phi}_W(\chi)$ is not, which facilitates the analysis of the asymptotic properties of $\tilde{E}[Wu(x^*)]$. The proof of our asymptotic normality result will in fact show that the distinction between $\tilde{\phi}_W(\chi)$ and $\hat{\phi}_W(\chi)$ becomes asymptotically irrelevant over an (expanding) interval of values of χ that is large enough to evaluate the integral in equation (5) with a numerical error negligible relative to the statistical noise. Not surprisingly, we indeed found that $\tilde{\phi}_W(\chi) = \hat{\phi}_W(\chi)$ over the range of values of χ used in our practical applications (Section 4). In any case, equation (31) *does not* require any user-specified sample-size dependent trimming and is thus simple to implement.

THEOREM 2: *Under Assumptions* 1 *through* 5, *for* $W_1, W_2 = 1, w_0, \ldots, w_L$ *and any* $u_1, u_2 \in \mathbb{S}$,

$$n^{1/2}\begin{bmatrix} \tilde{E}[W_1u_1(x^*)] - E[W_1u_1(x^*)] \\ \tilde{E}[W_2u_2(x^*)] - E[W_2u_2(x^*)] \end{bmatrix} \tag{32}$$

$$\xrightarrow{d} N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \Omega_{(W_1,u_1),(W_1,u_1)} & \Omega_{(W_1,u_1),(W_2,u_2)} \\ \Omega_{(W_2,u_2),(W_1,u_1)} & \Omega_{(W_2,u_2),(W_2,u_2)} \end{bmatrix}\right)$$

where, letting † denote complex conjugation,

$$\Omega_{(W_1,u_1),(W_2,u_2)} = \sum_{a=1,x,W_1} \sum_{b=1,x,W_2} \iint V_{ab}(\zeta,\xi) U_a^{W_1,u_1}(\zeta)\big(U_b^{W_2,u_2}(\xi)\big)^{\dagger} d\zeta d\xi \tag{33}$$

and similarly for $\Omega_{(W_1,u_1),(W_1,u_1)}$, $\Omega_{(W_2,u_2),(W_2,u_2)}$, *and where, for* $a,b=1,x,w_0,\ldots,w_L$, *and any* $\zeta,\xi\in\mathbb{R}$,

$$V_{ab}(\zeta,\xi) = \big(m_{ab}(\zeta-\xi) - m_a(\zeta)m_b(-\xi)\big), \tag{34}$$

$$m_{ab}(\zeta) = E[abe^{i\zeta z}], \quad m_a(\zeta) = E[ae^{i\zeta z}], \tag{35}$$

and where, for $a=1,x,w_0,\ldots,w_L$, *for* $W=1,w_0,\ldots,w_L$, *any* $u\in\mathbb{S}$ *and any* $\zeta\in\mathbb{R}$,

$$U_a^{W,u}(\zeta) = \begin{cases} -\dfrac{iB_{1,u}(\zeta)m_x(\zeta)}{(m_1(\zeta))^2} & \text{if } W=1 \text{ and } a=1, \\ \dfrac{iB_{1,u}(\zeta)}{m_1(\zeta)} & \text{if } W=1 \text{ and } a=x, \\ 0 & \text{if } W=1 \text{ and } a=w_j \text{ for some } j\in\{0,\ldots,L\}, \\ -\dfrac{iB_{w_l,u}(\zeta)m_x(\zeta)}{(m_1(\zeta))^2} - \dfrac{C_{w_l,u}(\zeta)}{m_1(\zeta)} & \\ \quad \text{if } W=w_l \text{ for some } l\in\{0,\ldots,L\} \text{ and } a=1, & \\ \dfrac{iB_{w_l,u}(\zeta)}{m_1(\zeta)} & \text{if } W=w_l \text{ for some } l\in\{0,\ldots,L\} \text{ and } a=x, \\ \dfrac{C_{1,u}(\zeta)}{m_1(\zeta)} & \text{if } W=w_l \text{ for some } l\in\{0,\ldots,L\} \text{ and } a=w_j \\ & \text{for some } j\in\{0,\ldots,L\}, \end{cases} \tag{36}$$

$$C_{W,u}(\zeta) = \frac{1}{2\pi}\mu_u(-\zeta)\phi_W(\zeta) \quad \text{for all } \zeta, \tag{37}$$

$$B_{W,u}(\zeta) = \begin{cases} \dfrac{1}{2\pi}\displaystyle\int_{\zeta}^{\infty} \mu_u(-\chi)\phi_W(\chi)d\chi & \text{for } \zeta\geq 0, \\ \dfrac{1}{2\pi}\displaystyle\int_{\zeta}^{-\infty} \mu_u(-\chi)\phi_W(\chi)d\chi & \text{for } \zeta<0, \end{cases} \tag{38}$$

for $\mu_u(\chi)$ *and* $\phi_W(\chi)$ *as in Theorem* 1.

The proof of this result, given in the Appendix, can be outlined as follows. (Throughout the proof, the fact that $\mu_u(\chi)$ may be a distribution in $\mathcal{D}_{s,\bar{\zeta}}$ rather than a simple function is handled by s repeated integrations by parts.) The estimated moments $\tilde{E}[Wu(x^*)] = \int \mu_u(-\chi)\tilde{\phi}_W(\chi)d\chi$ for $W = 1, w_0, \ldots, w_L$ and $u \in \mathbb{S}$ are expanded in powers of $(\hat{m}_a(\zeta) - m_a(\zeta))$ for $a = 1, x, w_0, \ldots, w_L$. The linear term of this expansion gives an expression for the influence function of the estimator, which is shown to have a finite variance, thus directly implying the asymptotic normality and root n consistency result. In order to justify this linearization, the higher-order terms are shown to be $o_p(n^{-1/2})$ through two techniques. For small values of the integration variable χ, we show that $\sup_{\zeta\in[-\tau_n,\tau_n]} |\hat{m}_a(\zeta) - m_a(\zeta)| \xrightarrow{p} 0$ sufficiently fast, for some sequence $\tau_n \to \infty$. This enables us to bound the higher order powers of $(\hat{m}_a(\zeta) - m_a(\zeta))$ over the expanding interval $[-\tau_n, \tau_n]$ in the stochastic expansion of $\int_{|\chi|\le\tau_n} \mu_u(-\chi)\tilde{\phi}_W(\chi)d\chi$. For large values of the integration variable ($|\chi| > \tau_n$), we show that even if $|\tilde{\phi}_W(\chi)|$ reached its maximum possible value ($2\hat{\alpha}_W$), the remainder $\int_{|\chi|>\tau_n} \mu_u(-\chi)\tilde{\phi}_W(\chi)d\chi$ is still negligible provided that $\int_{\tau_n}^{\infty} |\mu_u(\chi)|d\chi = o(n^{-1/2})$, a requirement implied by the regularity condition $u \in \mathbb{S}$. The cutoff τ_n is solely used as a device of proof and does *not* imply the need for a user-specified trimming parameter in the practical implementation of the estimator. It is the data-driven constraint imposed by equation (31) that circumvents trimming.

Theorem 2 not only provides the true asymptotic variance of the estimator, but also offers a way to estimate standard errors in a finite sample. Once $\phi_W(\chi)$ is replaced by its value given by equation (6), the expression for the asymptotic variance provided in Theorem 2 only involves moments of observable variables.

It may come as a surprise that our estimator can achieve root n consistency, given the well-known slow convergence problem encountered by nonparametric kernel-deconvolution estimators (Carroll and Hall (1988), Fan (1991), Fan and Truong (1993)). The crucial distinction lies in the fact that, in our estimator, the estimated characteristic function $\tilde{\phi}_1(\chi)$ is multiplied by a sample size-independent deterministic function $\mu_u(\chi)$ that downweights the noise in $\tilde{\phi}_1(\chi)$ for large χ and ensures that the influence function of the estimator has finite variance. In the case of nonparametric kernel-deconvolution estimators, the Fourier transform of the kernel (which plays the role of a sample size-*dependent* function $\mu_u(\chi)$) converges to a constant function as $n \to \infty$ and is therefore unable to downweight the noisy tail of $\tilde{\phi}_1(\chi)$ sufficiently to enable the definition of an influence function with a variance that remains bounded as $n \to \infty$.

Our regularity conditions imposed through Definition 2 specifically ensure that the rate of decay of $\mu_u(\chi)$ is sufficiently fast to counter the increase in the noise on the estimated $\tilde{\phi}_1(\chi)$ as $|\chi| \to \infty$. In the deconvolution literature (Carroll and Hall (1988), Fan (1991), Fan and Truong (1993)), regularity conditions are often specified in this fashion, that is, in terms of upper bounds on the rates of decay of the various Fourier transforms involved in an estimator. Such conditions are considered primitive, because the speed at which a Fourier transform decays to zero as χ goes to infinity is governed by the smoothness of the original function, as indicated by the following definition and associated theorem, shown in Lighthill (1962).

DEFINITION 4: A function $u : \mathbb{R} \mapsto \mathbb{R}$ is said to be *well behaved at infinity* if $\int_{|t|\geq T} |u(t) - u_\infty(t)|dt < \infty$ for some $T \in \mathbb{R}^+$ and some function $u_\infty(t)$ that can be written as a finite linear combination of products of functions of the form $|t|^c$, $\operatorname{sgn}(t)|t|^c$, $\ln|t|$, $\sin(ct)$, $\cos(ct)$ with $c \in \mathbb{R}^+$. (Note that a probability density is always well behaved at infinity.)

THEOREM 3: *If a function $u : \mathbb{R} \mapsto \mathbb{R}$ has d derivatives that are well behaved at infinity and absolutely integrable in every finite interval, then its Fourier transform, $\mu_u(\tau)$, is such that $|\mu_u(\tau)| = o(|\tau|^{-d})$ as $|\tau| \to \infty$.*

As seen in Definition 2, the constraint on the rate of decay of $\mu_u(\chi)$ is mainly[8] a function of the asymptotic behavior of $m_1(\zeta)$ as $|\zeta| \to \infty$: The faster $|m_1(\zeta)|$ decreases as $|\zeta| \to \infty$, the faster $\mu_u(\chi)$ needs to decay towards zero as $|\chi| \to \infty$ and the smoother $u(x^*)$ needs to be. As a result, the size of the set $\mathbb{S}$ decreases as the density of the measurement error Δz becomes smoother, because $1/m_1(\zeta)$ then diverges faster as $\zeta \to \infty$. In contrast, there are no smoothness requirements on the densities of Δx and $w_0, \ldots, w_L$. Therefore, models with a discrete dependent variable w_0 or with perfectly observed dummy regressors pose no particular problem. To conclude our remarks on the regularity conditions, let us give a few examples of models that do or do not satisfy the regularity conditions.

EXAMPLE 1: Let $\lceil k \rceil = \min\{l \in \mathbb{N} : l \geq k\}$. If Δz and x^* both have a χ^2 distribution with $d_{\Delta z}$ and d_{x^*} degrees of freedom, respectively, then $u \in \mathbb{S}$ if $u(x^*)$ has $\lceil (d_{\Delta z} + d_{x^*} + 5)/2 \rceil$ derivatives that are absolutely integrable in any finite interval and well behaved at infinity.

EXAMPLE 2 (Probit model): If Δz and x^* are normally distributed with variance $\sigma^2_{\Delta z}$ and $\sigma^2_{x^*}$, respectively, and $u(x^*)$ is an error function $u(x^*) = (2\pi c^2)^{-1/2} \int_{-\infty}^{x^*} \exp(-(\tilde{x}^*)^2/(2c^2))d\tilde{x}^*$, then $u \in \mathbb{S}$ if and only if $c^2 > \sigma^2_{\Delta z} + \sigma^2_{x^*}$.

Having in hand an asymptotically normal and root n consistent estimator of any moment of the form $E[Wu(x^*)]$, we are now ready to state the asymptotic

[8]The quantity $m_x(\chi)/m_1(\chi)$ also plays a role but, for most common distributions, it increases no faster than logarithmically in χ and thus has little effect on the size of the set $\mathbb{S}$. Moreover, it does not depend on the density of the measurement error because $m_x(\chi)/m_1(\chi) = d(\ln \phi_1(\chi))/d\chi$.

properties of the coefficients β of a least square regression with a nonlinear specification and with one mismeasured regressor.

DEFINITION 5: Let $\beta = D^{-1}C$ where $D = E[XX']$, $C = E[yX]$, and $X = [h_1(x^*), \ldots, h_M(x^*), w_1, \ldots, w_L]'$. Let $\hat{\beta} = \hat{D}^{-1}\hat{C}$ denote an estimator where $\hat{D}$ and $\hat{C}$ are estimated element-by-element using Definition 3.

ASSUMPTION 6: *The functions $h_k(x^*)$ and $h_k(x^*)h_{k'}(x^*)$ for $k, k' = 1, \ldots, M$, belong to $\mathbb{S}$.*

ASSUMPTION 7: *D and C are finite and D is invertible.*

THEOREM 4: *Under Assumptions* 1 *through* 7, $n^{1/2}(\hat{\beta} - \beta) \xrightarrow{d} N(0, P\Omega P')$, *where $P = [-(C'D^{-1}) \otimes D^{-1}, D^{-1}]$ and Ω is the asymptotic covariance matrix of* $[\mathrm{vec}(\hat{D})', \hat{C}']'$, *obtained from*[9] *equation* (33), *and where $\otimes$ denotes the Kronecker product while* $\mathrm{vec}(\cdot)$ *denotes the usual column vectorization of a matrix.*

This result is shown using the standard delta method, computing the derivative of the normal equation $\beta = D^{-1}C$ with respect to the elements of both *C and D*. A complete proof appears in the Appendix.

3. GENERAL EXTREMUM ESTIMATORS

3.1. *The Model*

The results presented so far handle the identification of moments of the form $E[Wu(x^*)]$ when x^* is one-dimensional, thereby addressing the problem of measurement error in the common case of least squares regressions.[10] The techniques introduced in the previous section can be extended to allow for multidimensional mismeasured variables, as well as moments that are nonlinear in all the variables and in the parameter vector. We consider a general class of extremum estimators

$$\hat{\beta} = \arg\max_{\beta \in \mathcal{B}} H\big(\tilde{E}[R(x^*, w, \beta)]\big), \tag{39}$$

where the random vector x^* takes values in $\mathbb{R}^K$, while the random vector w takes values in $\mathbb{R}^L$; $H : \mathbb{R}^J \mapsto \mathbb{R}$ is a general nonlinear function; $R : \mathbb{R}^K \times \mathbb{R}^L \times$

[9]The submatrix of Ω giving the covariances between $w_l w_{l'}$ and $w_{l''} w_{l'''}$ for $l, l', l'', l''' = 1, \ldots, L$ can be set to zero, as it has no effect on the asymptotic variance of $\hat{\beta}$, just as in ordinary least squares.

[10]Nonparametric kernel estimators can also be handled, by noting that they can be written as a ratio of moments that our technique can identify. This avenue is explored in Schennach (2001).

$\mathbb{R}^B \mapsto \mathbb{R}^J$ can be nonlinear in the parameter vector β, the vector of true unobserved variables x^* that are measured with error, and the variables w that are perfectly observed; $\mathcal{B}$ is a compact subset of $\mathbb{R}^B$; $J, K, L, B \in \mathbb{N}$; and $\tilde{E}[\cdot]$ denotes an estimator of $E[\cdot]$, to be defined below. The vector w can also include variables where a measurement error, if present, would have no impact on the estimation procedure, such as the dependent variable in a conventional least squares regression. This class of estimators is sufficiently rich to cover most generalized method of moment estimators and maximum likelihood estimators. Once again, two error-contaminated measurements $x = x^* + \Delta x$ and $z = x^* + \Delta z$ of the vector x^* are available, and are assumed to satisfy restrictions that are analogous to Assumption 1 in Section 2.

ASSUMPTION 8: *$E[\Delta x_k | x_k^*, \Delta z_k] = 0$ and Δz_k is independent from x^*, w, and $\Delta z_{k'}$ for all $k', k = 1, \ldots, K$ such that $k' \neq k$.*

This additional requirement that Δz_k and $\Delta z_{k'}$ for $k \neq k'$ be independent is convenient, because it enables us to correct for the measurement error on each regressor separately. Relaxing this assumption would result in a significant increase in the complexity of the estimator,[11] an alternative we shall not explore here. The requirement that $E[w|x^*, \Delta z] = E[w|x^*]$ in the least squares case needs to be strengthened to a requirement of independence between w and Δz_k, because we are considering functions that may be nonlinear in w.

3.2. *Derivation of the Estimator*

Identifying a general nonlinear extremum estimator involves the identification of moments of the form $E[u(x^*, w, \beta)]$, where $u(x^*, w, \beta)$ is any one of $R_j(x^*, w, \beta)$ for $j = 1, \ldots, J$, the elements of the vector of functions $R(x^*, w, \beta)$. This process can conceptually be subdivided into two steps. First, the known function $u(x^*, w, \beta)$ is expressed as a linear combination of basis functions chosen to facilitate the treatment of measurement error. Second, the presence of measurement error in each term of the linear combination is taken into account.

The linear combination of basis functions used to represent $u(x^*, w, \beta)$ is chosen to be of the form:

$$u(x^*, w, \beta) = \sum_{\chi} \sum_{\omega} \mu_u(\chi, \omega, \beta) d_{\chi}(x^*) b_{\omega}(w) \tag{40}$$

[11]In equation (6), the inner integral could be replaced by a path integral where χ and ζ are vectors: $E[e^{i\chi \cdot x^*}] = \exp(\int_{P(0,\chi)} (iE[xe^{i\zeta \cdot z}]/E[e^{i\zeta \cdot z}]) \cdot d\zeta)$, where $P(0, \chi)$ is some continuous path connecting 0 and χ. This extension would, however, require the assumption that $E[\Delta x_k | x^*, \Delta z] = 0$.

where the $\mu_u(\chi, \omega, \beta)$ are the weights of the linear combination, while $d_\chi(x^*)$ for all χ and $b_\omega(w)$ for all ω form, respectively, a basis for functions of x^* and a basis for functions of w. Such a basis for functions of x^* and w is said to be *separable*: We find a basis $\{d_\chi(x^*)\}$ for functions of x^* and a basis $\{b_\omega(w)\}$ for functions of w separately, and obtain a basis for functions of x^* and w by forming every product of the form $\{d_\chi(x^*)b_\omega(w)\}$. The use of a separable basis is important, because, as we will see, the presence of measurement error can be directly taken into account for any moment of the form $E[d_\chi(x^*)b_\omega(w)]$.

In order to be able to use the results derived in the previous section, the basis $d_\chi(x^*)$ is taken to be the Fourier basis ($d_\chi(x^*) = e^{-i\chi\cdot x^*}$). In contrast, there are no restrictions on the choice of the basis $\{b_\omega(w)\}$. The result of this first step is a set of weights $\mu_u(\chi, \omega, \beta)$:

DEFINITION 6: For a given function $u : \mathbb{R}^K \times \mathbb{R}^L \times \mathbb{R}^B \mapsto \mathbb{R}$, let the function $\mu_u(\chi, \omega, \beta)$ be such that

$$(41)\qquad u(x^*, w, \beta) = (2\pi)^{-K} \sum_{\omega\in\mathcal{W}} \int \cdots \int \mu_u(\chi, \omega, \beta) e^{-i\chi\cdot x^*} b_\omega(w)\, d\chi_1 \ldots d\chi_K,$$

where $\chi = (\chi_1, \ldots, \chi_K)' \in \mathbb{R}^K$ and $\{b_\omega(w)\}$ is some set of basis functions indexed by ω belonging to some given finite-dimensional index set $\mathcal{W}$.

All the information regarding $u(x^*, w, \beta)$ is embodied in the coefficients $\mu_u(\chi, \omega, \beta)$ for all χ, ω, and β. The sum over ω can, of course, be converted to an integral, if the basis for functions of w is chosen to be continuous instead of discrete. It is important to note that although the potentially infinite sum over the basis functions needs to be truncated in practice, this truncation is based on a numerical accuracy criterion, independent of sample size, because $u(x^*, w, \beta)$ is known. The asymptotic properties of the estimator are therefore unaffected. In a very common particular case, the coefficients $\mu_u(\chi, \omega, \beta)$ can be directly determined. If the basis $b_\omega(w)$ is chosen to be orthonormal with respect to some weighting function $F(w)$ (such as Hermite polynomials, which are orthonormal with respect to a Gaussian weighting function),

$$(42)\qquad \int b_\omega^\dagger(\tilde{w}) b_\nu(\tilde{w}) F(\tilde{w})\, d\tilde{w} = 1(\omega = \nu),$$

for $\omega, \nu \in \mathcal{W}$, where $1(\cdot)$ denotes an indicator function, then $\mu_u(\chi, \omega, \beta)$ is given by

$$(43)\qquad \mu_u(\chi, \omega, \beta) = \int \cdots \int \left(\int u(\tilde{x}^*, \tilde{w}, \beta) F(\tilde{w}) b_\omega^\dagger(\tilde{w})\, d\tilde{w} \right) e^{i\chi\cdot\tilde{x}^*} d\tilde{x}_1^* \ldots d\tilde{x}_K^*.$$

In other cases, such as when $u(x^*, w, \beta)$ is a polynomial in w and the basis $b_\omega(w)$ is itself a polynomial basis, the weights $\mu_u(\chi, \omega, \beta)$ can be found by inspection.

Once $\mu_u(\chi, \omega, \beta)$ has been determined, $E[u(x^*, w, \beta)]$ can be expressed in terms of observable variables using the following result.

ASSUMPTION 9: *$E[|x_k^*|]$, $E[|\Delta x_k|]$, are finite for $k = 1, \ldots, K$ and $E[|b_\omega(w)|]$ for all $\omega \in \mathcal{W}$ is uniformly bounded.*

THEOREM 5: *Under Assumptions 8 and 9, if $E[u(x^*, w, \beta)]$ exists, then*

$$
(44)\qquad E[u(x^*, w, \beta)] = (2\pi)^{-K} \sum_{\omega \in \mathcal{W}} \int \cdots \int \mu_u(-\chi, \omega, \beta)\phi_b(\chi, \omega)\, d\chi_1 \ldots d\chi_K
$$

where

$$
(45)\qquad \phi_b(\chi, \omega) = \frac{E[b_\omega(w)e^{i\chi\cdot z}]}{\prod_{k=1}^K E[e^{i\chi_k z_k}]} \left(\prod_{k=1}^K \exp\left(\int_0^{\chi_k} \frac{iE[x_k e^{i\zeta_k z_k}]}{E[e^{i\zeta_k z_k}]} d\zeta_k \right) \right)
$$

and where $\mu_u(\chi, \omega, \beta)$ is given in Definition 6.

PROOF: By the linearity of the integral and of the expectation operator, equation (41) implies that

$$
(46)\qquad E[u(x^*, w, \beta)] = (2\pi)^{-K} \sum_{\omega \in \mathcal{W}} \int \cdots \int \mu_u(-\chi, \omega, \beta) E\big[e^{i\chi\cdot x^*} b_\omega(w)\big] d\chi_1 \ldots d\chi_K,
$$

indicating that any expectation of the form $E[u(x^*, w, \beta)]$ can be obtained from the knowledge of expectations of the form $E[e^{i\chi\cdot x^*} b_\omega(w)]$. The second step of the identification procedure consists of expressing expectations of the form $E[e^{i\chi\cdot x^*} b_\omega(w)]$, which involve the unobserved variables x^*, solely in terms of the observable variables w, x, and z. By Assumption 8,

$$
\begin{aligned}
(47)\qquad E\big[b_\omega(w)e^{i\chi\cdot x^*}\big] &= \frac{E[b_\omega(w)e^{i\chi\cdot x^*}](\prod_{k=1}^K E[e^{i\chi_k \Delta z_k}])(\prod_{k=1}^K E[e^{i\chi_k x_k^*}])}{(\prod_{k=1}^K E[e^{i\chi_k x_k^*}])(\prod_{k=1}^K E[e^{i\chi_k \Delta z_k}])} \\
&= \frac{E[b_\omega(w)e^{i\chi\cdot z}](\prod_{k=1}^K E[e^{i\chi_k x_k^*}])}{\prod_{k=1}^K E[e^{i\chi_k z_k}]},
\end{aligned}
$$

where $E[e^{i\chi_k x_k^*}]$ can be expressed in terms of observable variables using equation (6) with $W = 1$. Substituting equation (47) into equation (46) yields the desired result. *Q.E.D.*

An estimator $\tilde{E}[u(x^*, w, \beta)]$ can be obtained as follows.

DEFINITION 7: Let $\hat{\phi}_b(\chi, \omega)$ be obtained by replacing all the expectations of observable variables by the corresponding sample moments in equation (45). Let $\tilde{\phi}_b(\chi, \omega) = \hat{\phi}_b(\chi, \omega)\min\{2\hat{\alpha}(\omega)/|\hat{\phi}_b(\chi, \omega)|, 1\}$, where $\hat{\alpha}(\omega)$ is the sample average estimating[12] $E[|b_\omega(w)|]$. Then $\tilde{E}[u(x^*, w, \beta)]$ is given by equation (44) with $\phi_b(\chi, \omega)$ replaced by $\tilde{\phi}_b(\chi, \omega)$.

3.3. *Asymptotic Properties*

The derivation of the asymptotic properties closely follows the derivation of the asymptotics of the least squares estimator just presented, so that we will focus our attention on differences.

ASSUMPTION 10: *$(x'_{\cdot j}, z'_{\cdot j}, w'_{\cdot j})'$ for $j = 1, \dots, n$ is an iid sample of the random variables $(x', z', w')'$, where $x_{\cdot j} \equiv (x_{1j}, \dots, x_{Kj})'$, $z_{\cdot j} \equiv (z_{1j}, \dots, z_{Kj})'$, and $w_{\cdot j} \equiv (w_{1j}, \dots, w_{Lj})$.*

ASSUMPTION 11: *$E[|x_k|^{2-j}|z_k|^j] < \infty$ and there exists a function $\tilde{b}(w)$ such that $|b_\omega(w)| \le \tilde{b}(w)$ for all $\omega \in \mathcal{W}$ and $E[|\tilde{b}(w)|^{2-j}|z_k|^j] < \infty$ for $k = 1, \dots, K$ and for $j = 0, 1$.*

ASSUMPTION 12: *$|E[e^{i\zeta_k z_k}]| > 0$ for any finite ζ_k, for $k = 1, \dots, K$.*

ASSUMPTION 13: *There exist a function $\bar{b}(w)$ (with $E[\bar{b}(w)] < \infty$) and a metric $\bar{d}(\cdot, \cdot)$ such that the metric space $(\mathcal{W}, \bar{d})$ is compact and such that $|b_\omega(w) - b_\nu(w)| \le \bar{b}(w)\bar{d}(\omega, \nu)$ for any $\omega, \nu \in \mathcal{W}$.*

While Assumptions 10, 11, and 12 are straightforward extensions of Assumptions 3, 4, and 5, Assumption 13 is a new requirement that imposes constraints on our choice of the basis functions $b_\omega(w)$, in order to ensure that the family of moments involving $b_\omega(w)$ converges in probability uniformly for all $\omega \in \mathcal{W}$. If only a finite number of basis functions $b_\omega(w)$ are needed to represent $u(x^*, w, \beta)$, Assumption 13 is automatically satisfied. The following set is used to further specify our regularity conditions.

DEFINITION 8: For a given set $\mathcal{C} \subseteq \mathbb{R}^B$, let $\mathbb{S}'_{\mathcal{C}}$ be the set of all functions $u : \mathbb{R}^K \times \mathbb{R}^L \times \mathcal{C} \mapsto \mathbb{R}$ such that[13]

$$(48) \qquad |\mu_u(\chi, \omega, \beta)| \le \bar{\mu}_0(\omega) \prod_{k=1}^{K} \bar{\mu}_k(\chi_k)$$

[12]That is, $\hat{\alpha}(\omega) = n^{-1}\sum_{j=1}^{n} |b_\omega(w_{\cdot j})|$, for $w_{\cdot j}$ as defined in Assumption 10.

[13]The inequality is to be understood as a dominance condition in the case of distributions. For instance, $A\delta(\chi) \le \delta(\chi)$ iff $A \le 1$.

for all $\beta \in \mathcal{C}$, where $\bar{\mu}_0 : \mathcal{W} \mapsto \mathbb{R}^+$ satisfies $\sum_{\omega \in \mathcal{W}} \bar{\mu}_0(\omega) < \infty$ and $\bar{\mu}_k$ satisfies the following two requirements. First, for some given $\bar{\zeta}$, $A \in \mathbb{R}^+$, and $\eta \in]0, 1[$,

$$\int_{\zeta_k}^{\infty} \bar{\mu}_k(\chi_k) d\chi_k \leq A(\lambda_k(\zeta_k))^{-1-\eta}, \tag{49}$$

for all $\zeta_k \geq \bar{\zeta}$ and for $k = 1, \dots, K$, where

$$\lambda_k(\zeta_k) = (1 + |\zeta_k|)\left(\sup_{\chi_k \in [-\zeta_k, \zeta_k]} \frac{|E[x_k e^{i\chi_k z_k}]|}{|E[e^{i\chi_k z_k}]|}\right)\left(\sup_{\chi_k \in [-\zeta_k, \zeta_k]} \left|E[e^{i\chi_k z_k}]\right|^{-1}\right). \tag{50}$$

Second, there exists $s \in \mathbb{N}$ such that $\bar{\mu}_k \in \mathcal{D}_{s,\bar{\zeta}}$ (from Definition 1), and $E[|a|^2 |z_k|^{2s}]$ exists for $a = 1, x_k, b_\omega(w)$, for $k = 1, \dots, K$ and $\forall \omega \in \mathcal{W}$.

Let $\mathbb{S}'_\emptyset$ denote the set of all functions satisfying the above constraints but that do not depend on β.

The form of the dominating condition of equation (48) enables us to specify regularity conditions separately for each dimension of χ, therefore permitting a straightforward multivariate extension of the regularity conditions imposed through Definition 2 in Section 2. A new requirement is the restriction on the dependence of the moments on ω, as imposed by equation (48). Without specifying the basis $\{b_\omega(w)\}$, one cannot directly relate this condition to the smoothness of the quantities involved. Fortunately, thanks to the fact that the basis $b_\omega(w)$ can be chosen, the dependence of $u(x^*, w, \beta)$ on w can often be represented by a small number of basis functions, so that the sum $\sum_{\omega \in \mathcal{W}} \bar{\mu}_0(\omega)$ reduces to a finite number of terms, whose summability is trivial to verify.

The following quantities are needed in order to state the asymptotic properties of the estimator.

DEFINITION 9: Let

$$\begin{aligned} S_0(\chi, \omega, x, w, z) &= b_\omega(w) e^{i\chi \cdot z}, \\ S_k(\chi, \omega, x, w, z) &= e^{i\chi_k z_k} \quad \text{for } k = 1, \dots, K, \\ S_{K+k}(\chi, \omega, x, w, z) &= x_k e^{i\chi_k z_k} \quad \text{for } k = 1, \dots, K, \end{aligned} \tag{51}$$

define $s_j(\chi, \omega) = E[S_j(\chi, \omega, x, w, z)]$ for $j = 0, \dots, 2K$, and let $\hat{s}_j(\chi, \omega)$ denote the corresponding sample average. Define, for any function $u : \mathbb{R}^K \times \mathbb{R}^L \times \mathbb{R}^B \mapsto \mathbb{R}$ and for $k = 1, \dots, K$,

$$U_{u,0}(\chi, \omega, \beta) = D_{u,\omega}(\chi, \beta), \tag{52}$$

$$\begin{aligned} U_{u,k}(\chi, \omega, \beta) = -&\left(\frac{iB_{u,k}(\chi_k, \beta) E[x_k e^{i\chi_k z_k}]}{(E[e^{i\chi_k z_k}])^2} + \frac{C_{u,k}(\chi_k, \beta)}{E[e^{i\chi_k z_k}]}\right) \\ &\times 1(\omega = \omega_0) \prod_{k' \neq k} \delta(\chi_{k'}), \end{aligned} \tag{53}$$

$$U_{u,K+k}(\chi,\omega,\beta)=\frac{iB_{u,k}(\chi_k,\beta)}{E[e^{i\chi_k z_k}]}1(\omega=\omega_0)\prod_{k'\neq k}\delta(\chi_{k'}), \tag{54}$$

where ω_0 is some fixed arbitrary element of $\mathcal{W}$ and where

$$B_{u,k}(\chi_k,\beta)=\begin{cases}(2\pi)^{-1}\displaystyle\int_{\chi_k}^{\infty}E\big[e^{i\zeta_k x_k^*}\mu_u(-\zeta_k,x_{-k}^*,w,\beta)\big]d\zeta_k\\ \quad\text{for } \chi_k\geq 0,\\ (2\pi)^{-1}\displaystyle\int_{\chi_k}^{-\infty}E\big[e^{i\zeta_k x_k^*}\mu_u(-\zeta_k,x_{-k}^*,w,\beta)\big]d\zeta_k\\ \quad\text{for } \chi_k<0,\end{cases} \tag{55}$$

$$C_{u,k}(\chi_k,\beta)=(2\pi)^{-1}E\big[e^{i\chi_k x_k^*}\mu_u(-\chi_k,x_{-k}^*,w,\beta)\big], \tag{56}$$

$$D_{u,\omega}(\chi,\beta)=(2\pi)^{-K}\mu_u(-\chi,\omega,\beta)\left(\frac{\prod_{k'=1}^{K}E[e^{i\chi_{k'}x_{k'}^*}]}{\prod_{k'=1}^{K}E[e^{i\chi_{k'}z_{k'}}]}\right), \tag{57}$$

$$\begin{aligned}&\mu_u(\chi_k,x_{-k}^*,w,\beta)\\ &\quad=\int u\big((x_1^*,\ldots,x_{k-1}^*,\tilde{x}_k^*,x_{k+1}^*,\ldots,x_K^*)',w,\beta\big)e^{i\chi_k\tilde{x}_k^*}d\tilde{x}_k^*.\end{aligned} \tag{58}$$

A few remarks are in order. The indicator function $1(\omega=\omega_0)$ and the delta functions[14] $\prod_{k'\neq k}\delta(\chi_k)$ are included in the definition of $U_{u,j}(\chi,\omega,\beta)$ merely to simplify the notation later on, allowing us to integrate over all variables ω, $\chi_1,\ldots,\chi_K$, regardless of whether a given moment depends on all variables or on only one of them. Also, note that the quantity $E[e^{i\chi_k x_k^*}\mu_u(-\chi_k,x_{-k}^*,w,\beta)]$, which enters the definition of $B_{u,k}(\chi_k,\beta)$ and $C_{u,k}(\chi_k,\beta)$, can be expressed in terms of observable variables through the following expression, obtained by substituting equation (41) into equation (58), multiplying by $e^{i\chi_k x_k^*}$, taking the expectation, and finally using Theorem 5:

$$\begin{aligned}&E\big[e^{i\chi_k x_k^*}\mu_u(-\chi_k,x_{-k}^*,w,\beta)\big]\\ &\quad=(2\pi)^{-K+1}\sum_{\omega\in\mathcal{W}}\int\cdots\int\mu_u(-\chi,\omega,\beta)\left(\frac{E[b_\omega(w)e^{i\chi\cdot z}]}{\prod_{k'=1}^{K}E[e^{i\chi_{k'}z_{k'}}]}\right)\\ &\qquad\times\left(\prod_{k'=1}^{K}\exp\left(\int_0^{\chi_{k'}}\frac{iE[x_{k'}e^{i\zeta_k z_{k'}}]}{E[e^{i\zeta_k z_{k'}}]}d\zeta_k\right)\right)d\chi_1\ldots d\chi_{k-1}d\chi_{k+1}\ldots d\chi_K.\end{aligned} \tag{59}$$

The asymptotic properties of the estimator given in Definition 7 can now be stated.

[14] A delta function $\delta(\chi_k)$ satisfies $\int\delta(\chi_k)f(\chi_k)d\chi_k=f(0)$ for any function $f(\chi_k)$.

THEOREM 6: *Under Assumptions* 8 *through* 13, *for any* $\gamma, \beta \in \mathbb{R}^B$, *any* $u_1 \in \mathbb{S}'_{\{\gamma\}}$, *and* $u_2 \in \mathbb{S}'_{\{\beta\}}$,

$$(60) \qquad n^{1/2}\begin{bmatrix} \tilde{E}[u_1(x^*, w, \gamma)] - E[u_1(x^*, w, \gamma)] \\ \tilde{E}[u_2(x^*, w, \beta)] - E[u_2(x^*, w, \beta)] \end{bmatrix} \xrightarrow{d} N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \Sigma_{u_1,u_1}(\gamma,\gamma) & \Sigma_{u_1,u_2}(\gamma,\beta) \\ \Sigma_{u_2,u_1}(\beta,\gamma) & \Sigma_{u_2,u_2}(\beta,\beta) \end{bmatrix}\right)$$

where all estimated expectations $\tilde{E}[\cdot]$ *are obtained by Definition* 7 *and where*

$$(61) \qquad \Sigma_{u_1,u_2}(\gamma,\beta) = \sum_{j=0}^{2K}\sum_{j'=0}^{2K}\sum_{\omega\in\mathcal{W}}\sum_{\nu\in\mathcal{W}}\int\cdots\int U_{u_1,j}(\chi,\omega,\gamma) \times \mathrm{Covar}\left[n^{1/2}\hat{s}_j(\chi,\omega), n^{1/2}\hat{s}_{j'}(\xi,\nu)\right] \times U^{\dagger}_{u_2,j'}(\xi,\nu,\beta)\,d\chi_1\ldots d\chi_K\,d\xi_1\ldots d\xi_K,$$

and similarly for $\Sigma_{u_1,u_1}(\gamma,\gamma)$, $\Sigma_{u_2,u_1}(\beta,\gamma)$, *and* $\Sigma_{u_2,u_2}(\beta,\beta)$, *where* $\chi = (\chi_1, \ldots, \chi_K)'$, $\xi = (\xi_1, \ldots, \xi_K)'$, *and* † *denotes complex conjugation*.

The proof, which is similar to the one of Theorem 2, is available upon request.

We are ultimately interested in the asymptotic properties of an extremum estimator defined by equation (39) using moments estimated through Definition 7. An important difference with the simple least square example of the previous section is the fact that we need to show uniform convergence in probability of moments of the form $\tilde{E}[u(x^*, w, \beta)]$ for all β in some compact set. The following definition and associated lemma, proven in the Appendix, are useful for this purpose.

DEFINITION 10: For a given set $\mathcal{C} \subseteq \mathbb{R}^B$, define the set $\mathcal{U}_{\mathcal{C}}$ of all functions $u : \mathbb{R}^K \times \mathbb{R}^L \times \mathcal{C} \mapsto \mathbb{R}$ such that $u \in \mathbb{S}'_{\mathcal{C}}$ and such that there exist $\bar{u} \in \mathbb{S}'_{\emptyset}$ and a metric $d(\cdot,\cdot)$ such that the metric space $(\mathcal{C}, d)$ is compact and such that, for all $\gamma, \beta \in \mathcal{C}$,

$$(62) \qquad \left|u(x^*, w, \gamma) - u(x^*, w, \beta)\right| \le \bar{u}(x^*, w)d(\gamma, \beta).$$

LEMMA 1: *Under Assumptions* 8 *through* 13, *if* $u \in \mathcal{U}_{\mathcal{C}}$, *then*

$$(63) \qquad \sup_{\beta\in\mathcal{C}}\left|\tilde{E}[u(x^*, w, \beta)] - E[u(x^*, w, \beta)]\right| \xrightarrow{p} 0.$$

We are now ready to state the asymptotic properties of the general extremum estimator defined by equation (39), making use of the results derived for arbitrary moments $E[u(x^*, w, \beta)]$. As usual, let subscripts denote specific elements of vectors or matrices.

ASSUMPTION 14: *$H(r(\beta))$, where $r(\beta) = E[R(x^*, w, \beta)]$, is uniquely maximized at $\beta = \beta^*$, the true value of the parameter vector, which lies in the interior of $\mathcal{B}$, a compact subset of $\mathbb{R}^B$.*

ASSUMPTION 15: *$H(\cdot)$ is twice continuously differentiable.*

ASSUMPTION 16: *For $k, l = 1, \ldots, B$, both $R^{(k)}(x^*, w, \beta) \equiv \partial R(x^*, w, \beta)/\partial\beta_k$ and $R^{(kl)}(x^*, w, \beta) \equiv \partial^2 R(x^*, w, \beta)/\partial\beta_k\partial\beta_l$ exist, are continuous (in β) for all x^* and all w, and are such that $\partial E[R(x^*, w, \beta)]/\partial\beta_k = E[R^{(k)}(x^*, w, \beta)] \equiv r^{(k)}(\beta)$, $\partial^2 E[R(x^*, w, \beta)]/\partial\beta_k\partial\beta_l = E[R^{(kl)}(x^*, w, \beta)] \equiv r^{(kl)}(\beta)$.*

ASSUMPTION 17: *The elements of $R(\cdot, \cdot, \cdot)$, denoted $R_j(\cdot, \cdot, \cdot)$ for $j = 1, \ldots, J$, belong to $\mathcal{U}_{\mathcal{B}}$ from Definition* 10.

DEFINITION 11: Let

$$q_l(g(\beta)) \equiv \sum_{j=1}^{J} \frac{\partial H(r(\beta))}{\partial r_j} \frac{\partial r_j(\beta)}{\partial \beta_l}, \tag{64}$$

where $g(\beta) \equiv E[g(x^*, w, \beta)] \in \mathbb{R}^{J(B+1)}$ and

$$\begin{aligned} g(x^*, w, \beta) \equiv \big(&R_1(x^*, w, \beta), \ldots, R_J(x^*, w, \beta), R_1^{(1)}(x^*, w, \beta), \ldots, \\ &R_J^{(1)}(x^*, w, \beta), \ldots, R_1^{(B)}(x^*, w, \beta), \ldots, R_J^{(B)}(x^*, w, \beta)\big)'. \end{aligned}$$

ASSUMPTION 18: *The matrix $(QG) \equiv Q(g(\beta^*))G(\beta^*)$ is nonsingular, where the elements of the matrices Q and G are given by $Q_{lj}(g) = \partial q_l(g)/\partial g_j$ and $G_{jk}(\beta) = \partial g_j(\beta)/\partial\beta_k$ for $k, l = 1, \ldots, B$, and $j = 1, \ldots, J$.*

ASSUMPTION 19: *$R_j^{(k)}(\cdot, \cdot, \cdot)$ and $R_j^{(kl)}(\cdot, \cdot, \cdot)$ belong to $\mathcal{U}_{\mathcal{N}}$ from Definition* 10 *for $k, l = 1, \ldots, B$, and $j = 1, \ldots, J$, where $\mathcal{N}$ is the closure of some open neighborhood of β^*.*

THEOREM 7: *Under Assumptions* 8 *through* 19,

$$n^{1/2}(\hat{\beta} - \beta^*) \xrightarrow{d} N\big(0, (QG)^{-1} Q \Gamma Q' (G'Q')^{-1}\big)$$

where the elements of matrix Γ are $\Gamma_{kl} = \Sigma_{g_k(\cdot,\cdot,\cdot), g_l(\cdot,\cdot,\cdot)}(\beta^*, \beta^*)$, as given by equation (61) *for $k, l = 1, \ldots, (J(B+1))$.*

The proof can be found in the Appendix. In light of the very general extension described in this section, it is apparent that a large class of models can be root n consistently estimated in the presence of measurement error in the regressors, given the availability of one repeated measurement of each mismeasured regressor. Examples include least squares type estimators (both linear and nonlinear in the parameter vector), discrete dependent variable models (such as logit and probit), limited dependent variable models (for instance truncated or censored regressions), duration models, and general minimum distance or extremum estimators.

4. APPLICATIONS

After discussing the practical implementation of the proposed estimator, this section illustrates its good finite-sample performances and provides an important application. A Monte Carlo study of a least squares model with a nonlinear specification illustrates the implementability and reliability of the proposed estimator. A practical application to Engel curves estimation is then presented.

4.1. *Computational Aspects*

While we focus here on estimators based on equation (5) to simplify the exposition, our analysis also applies to the more general estimator provided by equation (44), at the expense of additional notational details. In practice, the integration operations appearing in equation (5) are performed using standard numerical integration techniques, such as Simpson's rules, which involve sampling the integrand at a finite number of mesh points $\{0, \zeta_1, \zeta_2, \ldots, \zeta_N\}$. Of course, in order to ensure that the results are accurate, one needs to gradually increase the number of mesh points N until the value of the numerical integral fluctuates by less than a user-specified tolerance as the number of points is further increased. Because one of the integration bounds is infinite, both the density of points N/ζ_N and the maximum frequency ζ_N considered need to be gradually increased.

Thanks to two properties of characteristic functions, the dependence of $\hat{m}_a(\zeta)$ on ζ can indeed be accurately represented by considering only a finite set of values of ζ with a small loss of precision. Under mild conditions, characteristic functions (i) decay toward zero as frequency (ζ) goes to infinity and (ii) are smooth. Sufficient conditions that guarantee a fast rate of decay were presented earlier in Theorem 3 and explain why it is possible to truncate a characteristic function at a finite frequency ζ_N at the expense of a relatively small loss in precision. The frequency at which the characteristic function is truncated should not be confused with the bandwidth parameter in kernel estimators. The upper limit on frequency is a precision parameter that can be chosen independently of sample size. For a given sample size, the precision can be increased by increasing the cutoff frequency. The only limit is set by computer time. This is not so with kernel estimation, where the bandwidth (and thus the cutoff frequency) is a function of sample size and has a substantial impact on the statistical properties of the estimator.

The smoothness of a characteristic function or a Fourier transform such as $\phi_W(\zeta)$ can be quantified by noting that, by the Moment Theorem, $|d^k\phi_W(\zeta)/d\zeta^k| = |E[(x^*)^k We^{i\zeta x^*}]| \leq E[|W||x^*|^k]$. This bound on the derivative permits an accurate representation of the characteristic function, even if the mesh of values $0, \zeta_1, \dots, \zeta_N$ is relatively coarse.

In summary, the finiteness of the moments and the smoothness of a density guarantee that the number of values of ζ needed to represent the corresponding characteristic function with a given precision is kept under control. In our Monte Carlo study and our applications, we found that a mesh consisting of about 2000 points, which can be comfortably handled with today's computers, is typically sufficient to render the numerical noise negligible relative to the statistical noise.

The Fourier transforms were performed using the fast Fourier transforms (FFT) algorithm. Since this algorithm requires equispaced data, we computed the characteristic functions of z by applying the FFT algorithm to a histogram of the values of z_j. As the width of the bins was very narrow (we used from 1024 to 8192 bins), the loss in precision due to discretizing the data was negligible.

From a computational point of view, the smoothness and boundedness of the functions $u(x^*)$ is less of a concern, because these functions are known exactly and their Fourier transforms $\mu_u(\chi)$ could, in principle, be calculated analytically. This may be useful when $\mu_u(\chi)$ is a distribution rather than a simple function. In this case, the value of an integral of the form $\int \mu_u(\chi)\tilde{\phi}_W(\chi)d\chi$ can be obtained through integration by parts in order to handle any singularities in $\mu_u(\chi)$. Alternatively, a fully numerical approach would be to construct a sequence $\mu_{u,l}(\chi)$, $l = 1, 2, \dots$, of functions converging to the distribution $\mu_u(\chi)$. For instance, one can take $\mu_{u,l}(\chi) = \int_{-T_l}^{T_l} u(x^*)e^{i\chi x^*}dx^*$ for some sequence T_l diverging toward infinity. The integrals $I_l = \int \mu_{u,l}(\chi)\tilde{\phi}_W(\chi)d\chi$ are then evaluated numerically for each function $\mu_{u,l}(\chi)$ in the sequence, and the value of I_l for an l sufficiently large, so that I_l no longer depends on l substantially, is taken as an estimate of $\lim_{l\to\infty} I_l$. In our applications, we took this numerical approach.

4.2. *Monte Carlo Study*

To illustrate the potential for the proposed estimator to exhibit good finite-sample performances, this section reports the results of a simple Monte Carlo experiment. We consider estimation of the model

$$y = a + bx^* + c(e^{-x^*/2} - 1 + x^*/2) + \Delta y \tag{65}$$

using two error-contaminated measurements x, z of x^*. This specification is relevant to the estimation of Engel curves, as will be explained in more detail

TABLE I

MONTE CARLO SIMULATIONS[a] RESULTS

	OLS			IV			Fourier			
	Bias	σ	RMSE	Bias	σ	RMSE	Bias	σ	RMSE	ARMSE
a	0.103	0.026	0.106	0.063	0.034	0.072	0.003	0.045	0.045	0.059
	(0.003)	(0.002)	(0.003)	(0.003)	(0.002)	(0.003)	(0.004)	(0.003)	(0.003)	
b	−0.398	0.045	0.400	0.010	0.072	0.073	−0.005	0.072	0.072	0.078
	(0.005)	(0.003)	(0.005)	(0.007)	(0.005)	(0.005)	(0.007)	(0.005)	(0.005)	
c	−0.906	0.066	0.908	−0.796	0.154	0.811	−0.026	0.430	0.430	0.535
	(0.007)	(0.004)	(0.007)	(0.015)	(0.011)	(0.014)	(0.043)	(0.040)	(0.040)	

[a]The sample size is 1000 and the number of replications is 100. Standard deviations are indicated in parentheses.

below. The regressor $(e^{-x^*/2} - 1 + x^*/2)$ is used instead of $e^{-x^*/2}$ to reduce collinearity, in order to better illustrate the effect of measurement error on the linear and nonlinear parts of the specification.

Table I reports the results from 100 replications, each of which consists of a sample of 1000 observations drawn according to $x^* \sim N(0, 1)$, $\Delta x \sim N(0, (x^*)^4/4)$, $\Delta z \sim N(0, 1/4)$, $\Delta y \sim N(0, 1/4)$ and $a = 1$, $b = 1$, $c = 1$. For each replication, all variables are redrawn at random (including x^*). The number of significant digits we report reflects the precision of the numerical calculations of the Fourier-based estimator, while the standard deviations (in parentheses) provide the magnitude of the statistical noise of the simulation. The first estimator is ordinary least squares on the mismeasured regressors. The second one is the traditional IV estimator using nonlinear functions of the repeated measurement z as instruments (more specifically, 1, z, and $(e^{-z/2} - 1 + z/2)$). Note that in linear specifications, IV is consistent for both an instrument or a repeated measurement. The third estimator is our Fourier-based repeated measurement estimator. The same random numbers are used as an input for all estimators.

As is to be expected, our Fourier-based estimator leads to a substantial bias reduction relative to OLS. The IV estimator succeeds in reducing the bias on the coefficient of the linear regressor (b) but is unable to correct the bias on the coefficient of the nonlinear regressor (c). In contrast, the Fourier-based estimator essentially eliminates the bias for all coefficients. This elimination of the bias in the Fourier-based estimator comes at the price of a slight increase in the standard errors (σ). However, the overall root mean square error (RMSE), defined by $\sqrt{(\text{Bias})^2 + \sigma^2}$, is still much smaller than in the case of both OLS and IV.

This Monte Carlo experiment also shows that the asymptotic theory of the Fourier-based estimator performs well in practice, as indicated by the similarity between the asymptotic root mean square error (ARMSE) and the root mean square error obtain from Monte Carlo (RMSE). The ARMSE reported is the average over all replications.

4.3. *Application to Engel Curve Estimation*

The estimation of Engel curves has not only received great attention in the econometric literature, but Hausman, Newey, and Powell (1995) have also recently established the importance of measurement error in the estimation of nonlinear Engel curves. This section is meant to provide an empirical illustration of the estimator, rather than a thorough investigation of consumer behavior. For instance, we bypass discussions regarding the best specification for the form of Engel curves, and omit covariates that allow for demographic or regional price effects.

Our sample was taken from the Consumer Expenditure Survey from the United States Department of Labor (Nelson (1994)) and consists of all families that were interviewed during the first and second quarter of 1984. We focus on the specification preferred by Leser (1963):

$$\frac{y_{gj}}{X_j^*} = a + b\ln X_j^* + \frac{c}{X_j^*} + \epsilon_j, \tag{66}$$

where y_{gj} is consumer j's expenditure on commodity group g while X_j^* is the true total expenditure on all commodity groups, free of any reporting errors. Alternatively, X_j^* could be interpreted as permanent income.

To avoid a nonlinear measurement error on the left-hand side of equation (66), we multiply each side of the equation by X_j^*:

$$y_{gj} = aX_j^* + bX_j^*\ln X_j^* + c + X_j^*\epsilon_j. \tag{67}$$

Note that the transformed error term ($\epsilon_j' = X_j^*\epsilon_j$) still satisfies the required conditional mean requirement of Assumption 1. Since the error on total expenditure is usually assumed to be multiplicative, we let $x_j^* = \ln X_j^*$, so that a model expressed in terms of x_j^* exhibits additive measurement errors:

$$\begin{aligned} y_{gj} &= ae^{x_j^*} + bx_j^*e^{x_j^*} + c + \epsilon_j', \\ x_j &= x_j^* + \Delta x_j, \\ z_j &= x_j^* + \Delta z_j. \end{aligned} \tag{68}$$

The first measurement of expenditure, denoted by x_j, is taken to be the log expenditure reported in the current quarter, while the repeated measurement of expenditure, denoted by z_j, is taken to be the log expenditure reported in the next quarter. This choice allows for the error term ϵ_j' in Model (68) to be correlated with the measurement error on x_j. This is useful because the error on y_{gj} may indeed be correlated with the error on x_j since total expenditure is the sum over expenditures for each commodity group. The asymmetry in the assumptions made regarding the errors on each measurement thus turns out to be fruitful in this application, because reversing the role of x_j and z_j

TABLE II

ESTIMATED[a] ELASTICITIES AT EACH QUARTILE

	Fourier[b]			FGLS[c]			Hausman Test[d]
Percentile:	25%	50%	75%	25%	50%	75%	
Clothing	1.50	1.31	1.23	1.09	1.09	1.08	23.9
	(0.41)	(0.12)	(0.11)	(0.03)	(0.02)	(0.03)	
Entertainment	2.05	1.43	1.21	1.37	1.31	1.27	16.7
	(0.46)	(0.14)	(0.07)	(0.04)	(0.03)	(0.04)	
Food	0.80	0.77	0.73	0.75	0.68	0.62	44.3
	(0.09)	(0.04)	(0.04)	(0.01)	(0.01)	(0.01)	
Health	0.77	0.74	0.71	0.80	0.74	0.69	5.9
	(0.11)	(0.07)	(0.07)	(0.04)	(0.03)	(0.04)	
Transportation	0.33	1.00	1.41	1.65	1.60	1.52	73.2
	(0.62)	(0.28)	(0.18)	(0.04)	(0.03)	(0.03)	

[a]The number of observations is 3449. The specification is $y_{gj} = ae^{x_j^*} + bx_j^* e^{x_j^*} + c + \epsilon'_j$.

[b]Standard deviations are indicated in parentheses.

[c]The FGLS standard errors are included for the sake of completeness, although they are probably inconsistent, given the likely presence of measurement error.

[d]χ_3^2 statistic of a Hausman test of the absence of measurement error.

would result in a violation of the assumptions of the model. As discussed in Section 2, the assumptions of the model also allow for true log expenditure to vary over time, following a random walk shifted by a deterministic trend due, for instance, to business cycles, inflation, or seasonal variation in consumption.

As a point of reference, we also estimate Model (68) using feasible generalized least squares (FGLS). We use FGLS instead of OLS to account for the substantial heteroskedasticity present in the data.[15] The model of heteroskedasticity used for Model (68) is:

$$(69) \qquad \mathrm{Var}[\epsilon'_j] = \gamma_1 e^{x_j} + \gamma_2 e^{2x_j}.$$

The results of the estimation of Model (68) are shown in Table II. The number of significant digits reported reflects the numerical accuracy of the Fourier-based estimator. Since the coefficients of the demand equation are difficult to interpret, results are reported as elasticities (defined as $d(\ln y)/d(\ln X^*)$) evaluated at each quartile of total expenditure. Also shown are the FGLS estimation of each model and the results of a Hausman test (Hausman (1978)) verifying the hypothesis that the two estimates are the same.

The well-known "attenuation" phenomenon present in linear univariate regressions with measurement error can also be seen in our results. For all but

[15]It is important that the estimator used for comparison be efficient in the absence of measurement error. A Hausman test can then be performed to test whether it is important to account for measurement errors.

one commodity group (Transportation), the elasticities estimated taking into account measurement errors are higher than the elasticities estimated using FGLS.

The test statistic of a Hausman test of the hypothesis that measurement errors have no effect is distributed according to a χ^2 with 3 degrees of freedom under the null. As the critical value associated with a 95% level of confidence is 7.82, the hypothesis that measurement errors have no effects is rejected for all commodity groups, except for "Health." As an additional test of the presence of measurement error, it is useful to quantify the relative magnitude of the variances of x^*, Δx, and Δz. Under Assumption 1, it is straightforward to show that

$$\begin{aligned} &\operatorname{Var}[x^*] = \operatorname{Covar}[z, x], \\ &\operatorname{Var}[\Delta x] = \operatorname{Var}[x] - \operatorname{Covar}[z, x], \\ &\operatorname{Var}[\Delta z] = \operatorname{Var}[z] - \operatorname{Covar}[z, x]. \end{aligned} \tag{70}$$

Using these expressions, we find that

$$\sqrt{\frac{\operatorname{Var}[\Delta x_j]}{\operatorname{Var}[x_j^*]}} = 0.50 \quad \text{and} \quad \sqrt{\frac{\operatorname{Var}[\Delta z_j]}{\operatorname{Var}[x_j^*]}} = 0.44. \tag{71}$$

Hence, measurement errors can clearly not be neglected in the context of Engel curve estimation and our method presents a direct way to handle this issue, regardless of the functional form used.

5. CONCLUSION

This paper introduces a very general framework that permits the conversion of any member of a large class of estimators that are root n consistent and asymptotically normal in the absence of measurement error into estimators that are root n consistent and asymptotically normal in the presence of measurement errors, provided that a repeated measurement of each mismeasured regressor is available. This level of generality is achieved by developing a method to estimate the value of arbitrary moments involving unobserved "true" variables in terms of moments involving observed, but error contaminated variables. Since most estimators are already formulated in terms of moments, measurement error can be corrected for by simply "plugging in" the moments obtained by the proposed method into any conventional moment-based estimator. The proposed estimator exhibits all the desirable properties expected from any estimator: root n consistency, asymptotic normality, absence of distributional assumptions, and weak regularity conditions. A computer program implementing the estimator used in the applications of Section 4 is available upon request.

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APPENDIX

Throughout this appendix, for some given random variables a and z, we let $m_a(\zeta) = E[ae^{i\zeta z}]$ and $\hat{m}_a(\zeta) = n^{-1}\sum_{j=1}^{n} a_j e^{i\zeta z_j}$ where a_j, z_j denote sequences of iid random variables having the same distribution as the random variables a, z and where n is sample size. We allow a to be products of random variables, e.g. $m_{z^2x}(\zeta) = E[z^2 x e^{i\zeta z}]$. The following two lemmas collect easy-to-prove results that will be frequently used.

LEMMA 2: *For the random variable x as in Model* (1) *and under Assumption* 2, *for all $\zeta \in \mathbb{R}^+$ and for $\lambda(\zeta)$ given in Definition* 2, *the quantities* $(1+|\zeta|)$, $|m_x(\zeta)||m_1(\zeta)|^{-2}(1+|\zeta|)$, $(1+|\zeta|)|m_1(\zeta)|^{-1}$, $|m_1(\zeta)|^{-1}$, $(1+|m_x(\zeta)|/|m_1(\zeta)|)|m_1(\zeta)|^{-1}$, $\int_0^\zeta |m_1(\zeta)|^{-1}d\chi$, *and* $\int_0^\zeta |m_x(\zeta)||m_1(\zeta)|^{-2}d\chi$ *are all less than $A\lambda(\zeta)$ for some $A > 0$ independent of ζ.*

LEMMA 3: *Under Assumptions* 2 *and* 5, *for $\phi_W(\chi)$ and $\mu_u(\chi)$ as in Theorem* 1 *and for $\mathbb{S}$ and $\bar{\zeta}$ given in Definition* 2, *if $u \in \mathbb{S}$ then the integrals*

$$\int_0^\infty \int_\zeta^\infty 1(\chi \geq \tau)|\mu_u(\chi)||\phi_W(\chi)|d\chi |m_1(\zeta)|^{-1}d\zeta,$$

$$\int_0^\infty \int_\zeta^\infty 1(\chi \geq \tau)|\mu_u(\chi)||\phi_W(\chi)|d\chi |m_x(\zeta)||m_1(\zeta)|^{-2}d\zeta,$$

and

$$\int_\tau^\infty |\mu_u(\zeta)||\phi_W(\zeta)||m_1(\zeta)|^{-1}d\zeta$$

are bounded for $\tau \geq \bar{\zeta}$ and converge to zero as $\tau \to \infty$.

LEMMA 4: *Let z_j, a_j, and b_j be iid sequences of real-valued random variables* (*with $E[a_j^2] < \infty$ and $E[b_j^2] < \infty$*). *For any absolutely integrable nonrandom complex-valued functions $U_a(\zeta)$, $U_b(\zeta)$ of a real variable satisfying $U_a(\zeta) = U_a^\dagger(-\zeta)$ and $U_b(\zeta) = U_b^\dagger(-\zeta)$, where $\dagger$ denotes complex conjugation,*

$$n^{1/2}\begin{bmatrix} \int U_a(\zeta)(\hat{m}_a(\zeta) - m_a(\zeta))d\zeta \\ \int U_b(\zeta)(\hat{m}_b(\zeta) - m_b(\zeta))d\zeta \end{bmatrix} \xrightarrow{d} N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tilde{\Omega}_{aa} & \tilde{\Omega}_{ab} \\ \tilde{\Omega}_{ba} & \tilde{\Omega}_{bb} \end{bmatrix}\right) \tag{72}$$

where $\tilde{\Omega}_{ab} = \iint V_{ab}(\zeta, \xi)U_a(\zeta)U_b^\dagger(\xi)d\zeta d\xi$ and[16] *$V_{ab}(\zeta, \xi) = m_{ab}(\zeta - \xi) - m_a(\zeta)m_b(-\xi)$.*

PROOF: The proof proceeds by noting that $\int U_a(\zeta)(\hat{m}_a(\zeta) - m_a(\zeta))d\zeta$ is equal to an average of iid terms $n^{-1}\sum_{j=1}^{n}\psi_a(a_j, z_j)$, where the influence function

$$\psi_a(a_j, z_j) = a_j \int U_a(\zeta)e^{i\zeta z_j}d\zeta - \int U_a(\zeta)m_a(\zeta)d\zeta \tag{73}$$

[16] $\tilde{\Omega}_{aa}, \tilde{\Omega}_{ba}, \tilde{\Omega}_{bb}, V_{aa}, V_{ba}$, and V_{bb} are defined similarly.

is real-valued since $U_a(\zeta) = U_a^\dagger(-\zeta)$ by assumption and $m_a(\zeta) = m_a^\dagger(-\zeta)$ because it is the Fourier transform of a real quantity. The influence function also has a finite variance since

$$\text{(74)} \qquad \begin{aligned} E\big[|\psi_a(a_j, z_j)|^2\big] &\le E\left[\left|a_j \int U_a(\zeta) e^{i\zeta z_j} d\zeta\right|^2\right] + \left(\int U_a(\zeta) m_a(\zeta) d\zeta\right)^2 \\ &\le E\left[|a_j|^2 \left(\int |U_a(\zeta)||e^{i\zeta z_j}| d\zeta\right)^2\right] + \left(\int |U_a(\zeta)| E[|a_j|] d\zeta\right)^2 \\ &\le E\big[|a_j|^2\big]\left(\int |U_a(\zeta)| d\zeta\right)^2 + \big(E[|a_j|]\big)^2 \left(\int |U_a(\zeta)| d\zeta\right)^2 \\ &\le 2E\big[|a_j|^2\big]\left(\int |U_a(\zeta)| d\zeta\right)^2. \end{aligned}$$

It follows that $n^{-1}\sum_{j=1}^n \psi_a(a_j, z_j)$ is asymptotically normal. A similar result holds with "a" replaced by "b." The Cramér–Wold device and

$$\big|E\big[\psi_a(a_j, z_j)\psi_b(b_j, z_j)\big]\big|^2 \le E\big[|\psi_a(a_j, z_j)|^2\big] E\big[|\psi_b(b_j, z_j)|^2\big]$$

imply the joint normality of both sums. The asymptotic variance is given by

$$\text{(75)} \qquad \begin{aligned} \tilde{\Omega}_{ab} &= nE\left[\int U_a(\zeta)\big(\hat{m}_a(\zeta) - m_a(\zeta)\big) d\zeta \left(\int U_b(\xi)\big(\hat{m}_b(\xi) - m_b(\xi)\big) d\xi\right)^\dagger\right] \\ &= \iint nE\big[\big(\hat{m}_a(\zeta) - m_a(\zeta)\big)\big(\hat{m}_b(-\xi) - m_b(-\xi)\big)\big] U_a(\zeta) U_b^\dagger(\xi) d\zeta d\xi \end{aligned}$$

where the interchange of the integrals and the expectations is justified by the absolute integrability of $U_a(\zeta)$ and $U_b(\xi)$ and the finiteness of $E[a_j^2]$ and $E[b_j^2]$ and where

$$\text{(76)} \qquad \begin{aligned} & nE\big[\big(\hat{m}_a(\zeta) - m_a(\zeta)\big)\big(\hat{m}_b(-\xi) - m_b(-\xi)\big)\big] \\ &= E\big[\big(a_j \exp(i\zeta z_j) - m_a(\zeta)\big)\big(b_j \exp(-i\xi z_j) - m_b(-\xi)\big)\big] \\ &= E\big[a_j b_j \exp(i\zeta z_j - i\xi z_j)\big] - 2m_a(\zeta) m_b(-\xi) + m_a(\zeta) m_b(-\xi) \\ &= m_{ab}(\zeta - \xi) - m_a(\zeta) m_b(-\xi) = V_{ab}(\zeta, \xi). \end{aligned}$$

Q.E.D.

LEMMA 5: *Let z_j and a_j be iid sequences of real-valued random variables with $E[a_j^2] < \infty$. If a sequence of nonrandom functions $U_n(\zeta)$ satisfies $\lim_{n\to\infty} \int |U_n(\zeta)| d\zeta = 0$, then $\int U_n(\zeta)(\hat{m}_a(\zeta) - m_a(\zeta)) d\zeta = o_p(n^{-1/2})$.*

PROOF: The quantity of interest can be written as $n^{-1}\sum_{j=1}^n \psi_n(a_j, z_j)$ where $\psi_n(a_j, z_j) = \int U_n(\zeta)(a_j e^{i\zeta z_j} - m_a(\zeta)) d\zeta$. Along the lines of the proof of Lemma 4, we have that $E[|\psi_n(a_j, z_j)|^2] \le 2E[|a_j|^2](\int |U_n(\zeta)| d\zeta)^2 \to 0$, implying that $\mathrm{Var}[n^{-1}\sum_{j=1}^n \psi_n(a_j, z_j)] = n^{-1}E[|\psi_n(a_j, z_j)|^2] = o(n^{-1})$ which, by Chebyshev's inequality, implies the conclusion of the theorem. *Q.E.D.*

LEMMA 6: *If a_j and z_j are sequences of iid real-valued random variables such that $E[a_j^2] < \infty$ and $E[|a_j||z_j|] < \infty$, then, for any $\varepsilon > 0$ and $A, \kappa \ge 0$, $\sup_{\zeta \in [-An^\kappa, An^\kappa]} |\hat{m}_a(\zeta) - m_a(\zeta)| = o_p(n^{-1/2+\varepsilon})$.*

PROOF: Let $\delta\hat{m}_a(\zeta) = \hat{m}_a(\zeta) - m_a(\zeta)$. Following standard techniques, the rate of uniform convergence of $\delta\hat{m}_a(\zeta)$ is found by (i) obtaining the uniform convergence rate (in probability) over a discrete set $\{0, An^\kappa/N_n, 2An^\kappa/N_n, \ldots, An^\kappa\}$ of $N_n + 1$ equidistant mesh points and (ii) by showing uniform boundedness in probability of the derivative $d(\delta\hat{m}_a(\zeta))/d\zeta$ for all $\zeta \in \mathbb{R}$. The convergence rate over the discrete set is determined by employing Bernstein's inequality (Lemma 2.2.9 in van der Vaart and Wellner (1996)) while noting that $\max_{1\le j\le n}|a_j e^{i\zeta z_j}| \le n^{1/2+\varepsilon'/2}$ for any $\varepsilon' > 0$ with probability approaching one. This is the case because

$$\max_{1\le j\le n}\left|a_j e^{i\zeta z_j}\right| = \left(\max_{1\le j\le n}|a_j|^2\right)^{1/2} \le \left(\sum_{j=1}^{n}|a_j|^2\right)^{1/2} = n^{1/2}\left(n^{-1}\sum_{j=1}^{n}|a_j|^2\right)^{1/2}$$

where $n^{-1}\sum_{j=1}^{n}|a_j|^2 = O_p(1)$ since $E[a_j^2]$ is finite. The uniform boundedness in probability of $d(\delta\hat{m}_a(\zeta))/d\zeta$ is proven using the fact that $|d(\delta\hat{m}_a(\zeta))/d\zeta| \le n^{-1}\sum_{j=1}^{n}|a_j z_j| + E[|az|]$. Setting $N_n = A'(2E[|az|]+\rho)n^{\kappa+1/2-\varepsilon}$ for some $\rho, \varepsilon, A' > 0$ completes the proof. (A detailed proof is available upon request.) *Q.E.D.*

PROOF OF THEOREM 2: The proof proceeds by showing that estimated moments of the form $\tilde{E}[Wu(x^*)]$ for $W = 1, w_0, \ldots, w_L$ and $u \in \mathbb{S}$ admit an influence function representation, thus implying the joint asymptotic normality and root n consistency of any pair of such estimated moments. To this effect, we recall the following definitions. Let $\phi_W(\chi)$ be defined as in Theorem 1, and let $\hat{\phi}_W(\chi)$ and $\tilde{\phi}_W(\chi)$ be given by Definition 3. Let $\bar{\phi}_W(\chi)$ denote the linearization of $\hat{\phi}_W(\chi)$ with respect to all estimated moments $\hat{m}_a(\zeta)$ for $a = 1, x, w_0, \ldots, w_L$ in the vicinity of their true value $m_a(\zeta) = E[ae^{i\zeta z}]$:

$$\bar{\phi}_W(\chi) = \phi_W(\chi) + \phi_W(\chi)\int_0^{\chi}\left(\frac{i\delta\hat{m}_x(\zeta)}{m_1(\zeta)} - \frac{im_x(\zeta)\delta\hat{m}_1(\zeta)}{(m_1(\zeta))^2}\right)d\zeta + \phi_1(\chi)\left(\frac{\delta\hat{m}_W(\chi)}{m_1(\chi)} - \frac{m_W(\chi)\delta\hat{m}_1(\chi)}{(m_1(\chi))^2}\right) \tag{77}$$

where $\delta\hat{m}_a(\zeta) = \hat{m}_a(\zeta) - m_a(\zeta)$. We also decompose $u(x^*)$ and its Fourier transform $\mu_u(\chi)$ as

$$u(x^*) \equiv u^-(x^*) + u^+(x^*) \quad \text{and} \quad \mu_u(\chi) \equiv \mu_{u^-}(\chi) + \mu_{u^+}(\chi) \tag{78}$$

where $\mu_{u^-}(\chi) = 1(|\chi| < \bar{\zeta})\mu_u(\chi)$ and $\mu_{u^+}(\chi) = 1(|\chi| \ge \bar{\zeta})\mu_u(\chi)$ while $u^-(x^*)$ and $u^+(x^*)$ denote the inverse Fourier transforms of $\mu_{u^-}(\chi)$ and $\mu_{u^+}(\chi)$, respectively. The constant $\bar{\zeta}$ is as specified in Definition 2. Note that $u \in \mathbb{S} \Rightarrow u^-, u^+ \in \mathbb{S}$. By Definition 2, all the potential singularities in $\mu_u(\chi)$ are located in the interval $[-\bar{\zeta}, \bar{\zeta}]$ and are thus contained in $\mu_{u^-}(\chi)$, while $\mu_{u^+}(\chi)$ is defined on an infinite interval but is absolutely integrable. We then decompose the estimation error as follows:

$$\tilde{E}[Wu(x^*)] - E[Wu(x^*)] = \Psi^{W,u^+} + (1-\mathbf{1}_n)R_1^+ + \mathbf{1}_n R_1^+ + R_2^+ + \Psi^{W,u^-} + R_1^- \tag{79}$$

where $\mathbf{1}_n = 1(|\bar{\phi}_W(\chi)| \le 2\hat{\alpha}_W \ \forall |\chi| \le \tau_n)$ with $\hat{\alpha}_W = n^{-1}\sum_{j=1}^{n}|W_j|$, as in Definition 3, where

$$\Psi^{W,u^+} = \frac{1}{2\pi}\int \mu_{u^+}(-\chi)\big(\bar{\phi}_W(\chi) - \phi_W(\chi)\big)d\chi, \tag{80}$$

$$R_1^+ = \frac{1}{2\pi}\int_{|\chi|\le\tau_n} \mu_{u^+}(-\chi)\big(\tilde{\phi}_W(\chi) - \bar{\phi}_W(\chi)\big)d\chi, \tag{81}$$

$$R_2^+ = \frac{1}{2\pi}\int_{|\chi|>\tau_n} \mu_{u^+}(-\chi)\big(\tilde{\phi}_W(\chi) - \bar{\phi}_W(\chi)\big)d\chi, \tag{82}$$

$$\Psi^{W,u^-} = \frac{1}{2\pi}\int \mu_{u^-}(-\chi)\big(\bar{\phi}_W(\chi) - \phi_W(\chi)\big)d\chi, \tag{83}$$

$$R_1^- = \frac{1}{2\pi}\int \mu_{u^-}(-\chi)\big(\tilde{\phi}_W(\chi) - \bar{\phi}_W(\chi)\big)d\chi, \tag{84}$$

and where τ_n is a sample-size dependent cutoff such that $\lambda(\tau_n) = n^{1/2-\eta/4}$ for $\lambda(\zeta)$ and η as defined in Definition 2. Such a τ_n can always be found since $\lambda(\zeta)$ is increasing and $\lim_{\zeta\to\infty}\lambda(\zeta) = \infty$ by construction, while Assumptions 5 and 2 imply that $\lambda(\zeta)$ always exists and is continuous. We consider each term in turn. Making use of the definition of $\bar{\phi}_W(\chi)$, of the identity

$$\int_{-\infty}^{\infty}\int_0^{\chi} f(\chi,\zeta)\,d\zeta\,d\chi = \int_0^{\infty}\int_{\zeta}^{\infty} f(\chi,\zeta)\,d\chi\,d\zeta + \int_{-\infty}^{0}\int_{\zeta}^{-\infty} f(\chi,\zeta)\,d\chi\,d\zeta$$

for some function f, and of $\phi_1(\chi)m_W(\chi)/m_1(\chi) = \phi_W(\chi)$, we have

$$\begin{aligned}
\Psi^{W,u^+} &= \frac{1}{2\pi}\int_0^{\infty}\left(\int_{\zeta}^{\infty}\mu_{u^+}(-\chi)\phi_1(\chi)\frac{m_W(\chi)}{m_1(\chi)}d\chi\right)\left(\frac{i\delta\hat{m}_x(\zeta)}{m_1(\zeta)} - \frac{im_x(\zeta)\delta\hat{m}_1(\zeta)}{(m_1(\zeta))^2}\right)d\zeta \\
&\quad + \frac{1}{2\pi}\int_{-\infty}^{0}\left(\int_{\zeta}^{-\infty}\mu_{u^+}(-\chi)\phi_1(\chi)\frac{m_W(\chi)}{m_1(\chi)}d\chi\right) \\
&\quad \times\left(\frac{i\delta\hat{m}_x(\zeta)}{m_1(\zeta)} - \frac{im_x(\zeta)\delta\hat{m}_1(\zeta)}{(m_1(\zeta))^2}\right)d\zeta \\
&\quad + \frac{1}{2\pi}\int_{-\infty}^{\infty}\mu_{u^+}(-\chi)\phi_1(\chi)\left(\frac{\delta\hat{m}_W(\chi)}{m_1(\chi)} - \frac{m_W(\chi)\delta\hat{m}_1(\chi)}{(m_1(\chi))^2}\right)d\chi \\
&= \int\frac{iB_{W,u^+}(\zeta)}{m_1(\zeta)}\delta\hat{m}_W(\zeta)d\zeta - \int\frac{iB_{W,u^+}(\zeta)}{m_1(\zeta)}\frac{m_x(\zeta)}{m_1(\zeta)}\delta\hat{m}_1(\zeta)d\zeta \\
&\quad + \int\frac{C_{1,u^+}(\zeta)}{m_1(\zeta)}\delta\hat{m}_W(\zeta)d\zeta - \int\frac{C_{W,u^+}(\zeta)}{m_1(\zeta)}\delta\hat{m}_1(\zeta)d\zeta \\
&= \sum_{a=1,x,W}\int U_a^{W,u^+}(\zeta)\delta\hat{m}_a(\zeta)d\zeta
\end{aligned}\tag{85}$$

for $U_a^{W,u^+}(\zeta)$, $B_{W,u^+}(\zeta)$, and $C_{W,u^+}(\zeta)$ as defined in the statement of the theorem with u replaced by u^+. The interchange of the order of integration above is justified, as each of the integrands of the form $U_a^{W,u^+}(\zeta)\delta\hat{m}_a(\zeta)$ is absolutely integrable. (First, $\delta\hat{m}_a(\zeta)$ is bounded uniformly in ζ for any given sample since $|\delta\hat{m}_a(\zeta)| \le E[|a|] + n^{-1}\sum_{j=1}^{n}|a_j|$. Second, under Assumptions 2 and 5, Lemma 3 shows that $u^+\in\mathbb{S}$ implies that $\int|U_a^{W,u^+}(\zeta)|d\zeta < \infty$.)

This absolute integrability of $U_a^{W,u^+}(\zeta)$, along with Assumptions 3 and 4, enables the use of Lemma 4 with $U_a(\zeta) = U_a^{W,u^+}(\zeta)$ to show that the $\int U_a^{W,u^+}(\zeta)\delta\hat{m}_a(\zeta)d\zeta$ for $a = 1, x, W$ are jointly asymptotically normal and root n consistent, provided that $U_a^{W,u^+}(-\zeta) = (U_a^{W,u^+}(\zeta))^{\dagger}$, which can be readily verified. For instance, for $\zeta \ge \bar{\zeta}$,

$$\begin{aligned}
U_x^{W,u^+}(-\zeta) &= U_x^{W,u}(-\zeta) \\
&= iB_{W,u}(-\zeta)/m_1(-\zeta) = i\big(2\pi m_1(-\zeta)\big)^{-1}\int_{-\zeta}^{-\infty}\mu_u(-\chi)\phi_W(\chi)d\chi \\
&= -i\big(2\pi m_1(-\zeta)\big)^{-1}\int_{\zeta}^{\infty}\mu_u(\chi)\phi_W(-\chi)d\chi \\
&= (i)^{\dagger}\big(2\pi m_1^{\dagger}(\zeta)\big)^{-1}\int_{\zeta}^{\infty}\mu_u^{\dagger}(-\chi)\phi_W^{\dagger}(\chi)d\chi \\
&= \left(i\big(2\pi m_1(\zeta)\big)^{-1}\int_{\zeta}^{\infty}\mu_u(-\chi)\phi_W(\chi)d\chi\right)^{\dagger} = \big(U_x^{W,u^+}(\zeta)\big)^{\dagger},
\end{aligned}$$

where we have used the definition of $\mu_{u^+}(\chi)$ (from equation (78)) and the fact that $m_1(\zeta)$, $\mu_u(\chi)$, and $\phi_W(\chi)$ are Fourier transforms of real-valued functions, implying that $m_1(\zeta) = m_1^\dagger(-\zeta)$, etc. Similar derivations hold for the remaining $U_a^{W,u^+}(\zeta)$ for all $\zeta \in \mathbb{R}$. We then conclude that Ψ^{W,u^+} is asymptotically normal and root n consistent since the $\int U_a^{W,u^+}(\zeta)\delta\hat{m}_a(\zeta)d\zeta$ for $a = 1, x, W$ jointly are.

We will now bound the remainder terms $(1 - \mathbf{1}_n)R_1^+$, $\mathbf{1}_n R_1^+$, and R_2^+ of equation (79) by making repeated use of the bounds involving $\lambda(\tau_n)$ provided by Lemma 2 under Assumption 2 and by the following upper bound, given by Lemma 6 under Assumptions 3 and 4, for $\varepsilon = \eta/8$ with η as in Definition 2,

$$\text{(86)} \qquad \hat{M}_n \equiv \sup_{|\chi| \le \tau_n} \max_{a \in \{1, x, w_0, \ldots, w_L\}} |\delta\hat{m}_a(\chi)| = o_p(n^{\eta/8 - 1/2}),$$

provided that τ_n increases no faster than some power of n (which is the case because $\lambda(\zeta)$ increases faster than linearly and $\lambda(\tau_n) = n^{1/2 - \eta/4}$). Once each remainder term is bounded in terms of $\lambda(\tau_n)$ and $\hat{M}_n$, we make use of the facts that $\lambda(\tau_n)\hat{M}_n = n^{1/2-\eta/4} o_p(n^{\eta/8-1/2}) = o_p(1)$ and $\lambda(\tau_n)\hat{M}_n^2 = n^{1/2-\eta/4} o_p(n^{\eta/4-1}) = o_p(n^{-1/2})$ to show that all the remainder terms are $o_p(n^{-1/2})$.

Consider the term $(1 - \mathbf{1}_n)R_1^+$ of equation (79). We will now show that $(1 - \mathbf{1}_n) = 0$ with probability approaching 1 (w.p.a.1). The quantity $\bar{\phi}_W(\chi) \xrightarrow{p} \phi_W(\chi)$ uniformly over $[-\tau_n, \tau_n]$ since, by equation (77),

$$\text{(87)} \qquad \begin{aligned} \left|\bar{\phi}_W(\chi) - \phi_W(\chi)\right| &\le |\phi_W(\chi)| \int_0^\chi \frac{|\delta\hat{m}_x(\zeta)|}{|m_1(\zeta)|} d\zeta + |\phi_W(\chi)| \int_0^\chi \frac{|m_x(\zeta)|}{|m_1(\zeta)|^2} |\delta\hat{m}_1(\zeta)| d\zeta \\ &\quad + |\phi_1(\chi)| \frac{|\delta\hat{m}_W(\chi)|}{|m_1(\chi)|} + |\phi_W(\chi)| \frac{|\delta\hat{m}_1(\chi)|}{|m_1(\chi)|} \\ &\le A\big(|\phi_W(\chi)| + |\phi_1(\chi)|\big)\lambda(\chi)\hat{M}_n \\ &\le A\big(E[|W|] + 1\big)\lambda(\tau_n)\hat{M}_n = o_p(1) \end{aligned}$$

for $\chi \in [-\tau_n, \tau_n]$ and some $A > 0$. Since we have established that $\bar{\phi}_W(\chi) \xrightarrow{p} \phi_W(\chi)$ uniformly for $\chi \in [-\tau_n, \tau_n]$ and since $|\phi_W(\chi)| \le E[|W|] < 2E[|W|] \equiv 2\alpha_W$, it follows that $|\bar{\phi}_W(\chi)| < 2\alpha_W$ for all $\chi \in [-\tau_n, \tau_n]$ w.p.a.1. Since $\hat{\alpha}_W \xrightarrow{p} \alpha_W$ by the Kolmogorov law of large numbers, it follows that $|\bar{\phi}_W(\chi)| \le 2\hat{\alpha}_W$ for all $\chi \in [-\tau_n, \tau_n]$ w.p.a.1. Hence $\Pr[1 - \mathbf{1}_n \ne 0] \to 0$, which implies that $\Pr[(1 - \mathbf{1}_n)|R_1^+| \ne 0] \to 0$, whatever the value of R_1^+ may be, implying $\Pr[(1 - \mathbf{1}_n)|R_1^+| > n^{-1/2}] \to 0$ and $(1 - \mathbf{1}_n)R_1^+ = o_p(n^{-1/2})$.

Consider the term $\mathbf{1}_n R_1^+$ of equation (79). Whenever $\mathbf{1}_n = 1$, we have that $|\bar{\phi}_W(\chi)| \le 2\hat{\alpha}_W$ and the "constrained" $\tilde{\phi}_W(\chi)$ (which, by construction, is also such that $|\tilde{\phi}_W(\chi)| \le 2\hat{\alpha}_W$) is thus necessarily closer to $\bar{\phi}_W(\chi)$ than the "unconstrained" $\hat{\phi}_W(\chi)$:

$$\text{(88)} \qquad \begin{aligned} 2\pi\left|\mathbf{1}_n R_1^+\right| &\le \mathbf{1}_n \int_{|\chi| \le \tau_n} |\mu_{u^+}(-\chi)| \left|\tilde{\phi}_W(\chi) - \bar{\phi}_W(\chi)\right| d\chi \\ &\le \mathbf{1}_n \int_{|\chi| \le \tau_n} |\mu_{u^+}(-\chi)| \left|\hat{\phi}_W(\chi) - \bar{\phi}_W(\chi)\right| d\chi \\ &\le \int_{|\chi| \le \tau_n} |\mu_{u^+}(-\chi)| \left|\hat{\phi}_W(\chi) - \bar{\phi}_W(\chi)\right| d\chi \\ &= 2\int_0^{\tau_n} |\mu_{u^+}(-\chi)| \left|\hat{\phi}_W(\chi) - \bar{\phi}_W(\chi)\right| d\chi. \end{aligned}$$

This remainder of the linearization of $\hat{\phi}_W(\chi)$ is bounded using two standard expansions. First,

$$\frac{m_a(\zeta)+\delta\hat{m}_a(\zeta)}{m_1(\zeta)+\delta\hat{m}_1(\zeta)} = q_a(\zeta)+\delta\hat{q}_a(\zeta) \tag{89}$$

where $q_a(\zeta)=m_a(\zeta)/m_1(\zeta)$ and $\delta\hat{q}_a(\zeta)=\delta_1\hat{q}_a(\zeta)+\delta_2\hat{q}_a(\zeta)$ for $a=1,x,w_0,\dots,w_L$, where

$$\delta_1\hat{q}_a(\zeta)=\frac{\delta\hat{m}_a(\zeta)}{m_1(\zeta)}-\frac{m_a(\zeta)\delta\hat{m}_1(\zeta)}{(m_1(\zeta))^2}, \tag{90}$$

$$\delta_2\hat{q}_a(\zeta)=\frac{m_a(\zeta)}{m_1(\zeta)}\left(\frac{\delta\hat{m}_1(\zeta)}{m_1(\zeta)}\right)^2\left(1+\frac{\delta\hat{m}_1(\zeta)}{m_1(\zeta)}\right)^{-1}-\frac{\delta\hat{m}_a(\zeta)}{m_1(\zeta)}\frac{\delta\hat{m}_1(\zeta)}{m_1(\zeta)}\left(1+\frac{\delta\hat{m}_1(\zeta)}{m_1(\zeta)}\right)^{-1}. \tag{91}$$

Second, for $Q_x(\chi)=\int_0^\chi (im_x(\zeta)/m_1(\zeta))d\zeta$,

$$\delta\hat{Q}_x(\chi)=\int_0^\chi \big(i\hat{m}_x(\zeta)/\hat{m}_1(\zeta)\big)d\zeta-\int_0^\chi \big(im_x(\zeta)/m_1(\zeta)\big)d\zeta$$

and some $|\delta\bar{Q}_x(\chi)|\le|\delta\hat{Q}_x(\chi)|$,

$$\exp\big(Q_x(\chi)+\delta\hat{Q}_x(\chi)\big)=\exp\big(Q_x(\chi)\big)\left(1+\delta\hat{Q}_x(\chi)+\frac{1}{2}\exp\big(\delta\bar{Q}_x(\chi)\big)\big(\delta\hat{Q}_x(\chi)\big)^2\right). \tag{92}$$

By substituting expansions (89) and (92) into the identity

$$\hat{\phi}_W(\chi)=\big(\hat{m}_W(\chi)/\hat{m}_1(\chi)\big)\exp\left(\int_0^\chi \big(i\hat{m}_x(\zeta)/\hat{m}_1(\zeta)\big)d\zeta\right),$$

removing the terms linear in one of the $\delta\hat{m}_a(\zeta)$, and substituting the resulting expansion for $(\hat{\phi}_W(\chi)-\bar{\phi}_W(\chi))$ in equation (88), we find that $\mathbf{1}_nR_1^+$ can be decomposed as $|\mathbf{1}_nR_1^+|\le \pi^{-1}\sum_{j=1}^7 R_{1j}^+$ where

$$R_{11}^+=\int_0^{\tau_n}|\mu_{u^+}(-\chi)||\delta_1\hat{q}_W(\chi)||\phi_1(\chi)|\left(\int_0^\chi|\delta_1\hat{q}_x(\zeta)|d\zeta\right)d\chi, \tag{93}$$

$$R_{12}^+=\int_0^{\tau_n}|\mu_{u^+}(-\chi)||\delta_2\hat{q}_W(\chi)||\phi_1(\chi)|d\chi, \tag{94}$$

$$R_{13}^+=\int_0^{\tau_n}|\mu_{u^+}(-\chi)||\delta_2\hat{q}_W(\chi)||\phi_1(\chi)|\int_0^\chi|\delta_1\hat{q}_x(\zeta)|d\zeta d\chi, \tag{95}$$

$$R_{14}^+=\int_0^{\tau_n}|\mu_{u^+}(-\chi)||q_W(\chi)||\phi_1(\chi)|\int_0^\chi|\delta_2\hat{q}_x(\zeta)|d\zeta d\chi, \tag{96}$$

$$R_{15}^+=\int_0^{\tau_n}|\mu_{u^+}(-\chi)||\delta\hat{q}_W(\chi)||\phi_1(\chi)|\int_0^\chi|\delta_2\hat{q}_x(\zeta)|d\zeta d\chi, \tag{97}$$

$$R_{16}^+=\int_0^{\tau_n}|\mu_{u^+}(-\chi)||q_W(\chi)||\phi_1(\chi)|\frac{1}{2}\exp\big(|\delta\bar{Q}_x(\chi)|\big)\left(\int_0^\chi|\delta\hat{q}_x(\zeta)|d\zeta\right)^2d\chi, \tag{98}$$

$$R_{17}^+=\int_0^{\tau_n}|\mu_{u^+}(-\chi)||\delta\hat{q}_W(\chi)||\phi_1(\chi)|\frac{1}{2}\exp\big(|\delta\bar{Q}_x(\chi)|\big)\left(\int_0^\chi|\delta\hat{q}_x(\zeta)|d\zeta\right)^2d\chi. \tag{99}$$

These terms can then be bounded as follows:

$$R_{11}^+\le\int_0^{\tau_n}|\mu_{u^+}(-\chi)|\left(\frac{|\delta\hat{m}_W(\chi)|}{|m_1(\chi)|}+\frac{|m_W(\chi)||\delta\hat{m}_1(\chi)|}{|m_1(\chi)|^2}\right) \tag{100}$$

$$
\begin{aligned}
&\times|\phi_1(\chi)|\left(\int_0^{\chi}|\delta_1\hat{q}_x(\zeta)|d\zeta\right)d\chi\\
&\le A\lambda(\tau_n)\hat{M}_n\int_0^{\tau_n}|\mu_{u^+}(-\chi)|\left(1+\frac{|m_W(\chi)|}{|m_1(\chi)|}\right)|\phi_1(\chi)|\left(\int_0^{\chi}|\delta_1\hat{q}_x(\zeta)|d\zeta\right)d\chi\\
&\le A\lambda(\tau_n)\hat{M}_n\int_0^{\infty}\int_{\zeta}^{\infty}|\mu_{u^+}(-\chi)|\left(1+\frac{|m_W(\chi)|}{|m_1(\chi)|}\right)|\phi_1(\chi)|d\chi|\delta_1\hat{q}_x(\zeta)|d\zeta\\
&= A\lambda(\tau_n)\hat{M}_n\int_0^{\infty}\int_{\zeta}^{\infty}|\mu_{u^+}(-\chi)|\big(|\phi_1(\chi)|+|\phi_W(\chi)|\big)d\chi|\delta_1\hat{q}_x(\zeta)|d\zeta\\
&= A\lambda(\tau_n)\hat{M}_n\int_0^{\infty}\int_{\zeta}^{\infty}|\mu_{u^+}(-\chi)|\big(|\phi_1(\chi)|+|\phi_W(\chi)|\big)d\chi\\
&\quad\times\left(\frac{1}{|m_1(\zeta)|}+|m_x(\zeta)||m_1(\zeta)|^{-2}\right)\hat{M}_n d\zeta\\
&= A\lambda(\tau_n)\hat{M}_n^2\int_0^{\infty}\int_{\zeta}^{\infty}1(|\chi|\ge\bar{\zeta})|\mu_u(-\chi)|\big(|\phi_1(\chi)|+|\phi_W(\chi)|\big)d\chi\\
&\quad\times\left(\frac{1}{|m_1(\zeta)|}+|m_x(\zeta)||m_1(\zeta)|^{-2}\right)d\zeta\\
&= A'\lambda(\tau_n)\hat{M}_n^2=o_p(n^{-1/2})\quad\text{for some } A, A'>0,
\end{aligned}
$$

where we have used, in turn, (i) equation (90), (ii) equation (86) and $|m_1(\zeta)|^{-1}\le A\lambda(\zeta)$ from Lemma 2, (iii) the fact that

$$\int_0^{\tau_n}\int_0^{\chi}f(\chi,\zeta)d\zeta d\chi\le\int_0^{\infty}\int_0^{\chi}f(\chi,\zeta)d\zeta d\chi=\int_0^{\infty}\int_{\zeta}^{\infty}f(\chi,\zeta)d\chi d\zeta$$

for any positive function $f(\chi,\zeta)$, (iv) $|\phi_1(\chi)||m_W(\zeta)|/|m_1(\zeta)|=|\phi_W(\chi)|$, (v) equations (90) and (86), (vi) $\mu_{u^+}(-\chi)=1(|\chi|\ge\bar{\zeta})\mu_u(-\chi)$, (vii) Lemma 3. A similar derivation (available upon request) yields that $R_{1j}^+=o_p(n^{-1/2})$ for $j=2,\ldots,7$.

Next, we consider the R_2^+ term of equation (79) and decompose it as $R_2^+=R_{21}^++R_{22}^+$ where $R_{21}^+=(2\pi)^{-1}\int_{|\chi|>\tau_n}\mu_{u^+}(-\chi)(\tilde{\phi}_W(\chi)-\phi_W(\chi))d\chi$ and $R_{22}^+=-(2\pi)^{-1}\int_{|\chi|>\tau_n}\mu_{u^+}(-\chi)(\bar{\phi}_W(\chi)-\phi_W(\chi))d\chi$. Then,

$$
\begin{aligned}
(101)\qquad |R_{21}^+|&\le\frac{1}{2\pi}\int_{|\chi|>\tau_n}|\mu_{u^+}(-\chi)|\big(|\tilde{\phi}_W(\chi)|+|\phi_W(\chi)|\big)d\chi\\
&\le\frac{1}{\pi}(\hat{\alpha}_W+\alpha_W)\int_{|\chi|>\tau_n}|\mu_{u^+}(\chi)|d\chi\\
&\le\frac{A}{\pi}(\hat{\alpha}_W+\alpha_W)(\lambda(\tau_n))^{-1-\eta}\quad\text{for some } A>0 \text{ by } u\in\mathbb{S}\Rightarrow u^+\in\mathbb{S}\\
&=O_p(1)(n^{1/2-\eta/4})^{-1-\eta}\quad\text{by definition of }\tau_n\\
&=O_p(n^{-1/2-\eta/4+\eta^2/4})=o_p(n^{-1/2})\quad\text{since }\eta\in\,]0,1[.
\end{aligned}
$$

Following the same steps as in equation (85), $-R_{22}^+=\sum_{a=1,x,W}\int U_a^{W,v_n}(\zeta)\delta\hat{m}_a(\zeta)d\zeta$ where $v_n(\cdot)$ is the inverse Fourier transform of $\mu_u(\chi)1(|\chi|\ge\tau_n)$. By Assumptions 2 and 5 and Lemma 3, $\lim_{n\to\infty}\int|U_a^{W,v_n}(\zeta)|d\zeta=0$ and it follows, by Assumptions 3 and 4 and Lemma 5, that $\sum_{a=1,x,W}\int U_a^{W,v_n}(\zeta)\delta\hat{m}_a(\zeta)d\zeta=o_p(n^{-1/2})$.

Now, consider the terms $\Psi^{W,u^-} + R_1^- = (2\pi)^{-1}\int_{-\bar{\zeta}}^{\bar{\zeta}} \mu_{u^-}(-\chi)(\tilde{\phi}_W(\chi) - \phi_W(\chi))d\chi$ in equation (79). Since the techniques used parallel the ones used for the Ψ^{W,u^+} and R_1^+ terms above, we will only outline the steps while emphasizing the differences.

We first note that $\tilde{\phi}_W(\chi) = \hat{\phi}_W(\chi)$ for $\chi \in [-\bar{\zeta}, \bar{\zeta}]$ w.p.a.1 since (i) it can be shown that $\hat{\phi}_W(\chi) \xrightarrow{p} \phi_W(\chi)$ uniformly for $\chi \in [-\bar{\zeta}, \bar{\zeta}]$, (ii) $\hat{\alpha}_W \xrightarrow{p} \alpha_W$, as was shown earlier, and (iii) $|\phi_W(\chi)| \le \alpha_W < 2\alpha_W$, implying that the constraint in the definition of $\tilde{\phi}_W(\chi)$(equation (31)) is not binding over the fixed interval $[-\bar{\zeta}, \bar{\zeta}]$ w.p.a.1. Claim (i) follows from $\hat{m}_a(\zeta) \xrightarrow{p} m_a(\zeta)$ uniformly over the fixed interval $[-\bar{\zeta}, \bar{\zeta}]$ by equation (86), and from the fact that $\hat{\phi}_W(\chi)$ for $\chi \in [-\bar{\zeta}, \bar{\zeta}]$ is a continuous mapping of the $\hat{m}_a(\zeta)$ for $\zeta \in [-\bar{\zeta}, \bar{\zeta}]$ in the supremum norm (by expansions (89) and (92), the fact that $|m_a(\zeta)| \le E[|a|] < \infty$ for $a = 1, x, w_0, \dots, w_L$ by Assumption 2, and $|m_1(\zeta)|^{-1} \le A\lambda(\bar{\zeta}) < \infty$ for some $A > 0$ by Lemma 2). We can thus focus on $(2\pi)^{-1}\int_{-\bar{\zeta}}^{\bar{\zeta}} \mu_{u^-}(-\chi)(\hat{\phi}_W(\chi) - \phi_W(\chi))d\chi$ instead. The singularities in $\mu_{u^-}(\chi)$ can be handled by integrations by parts, noting that the boundary terms are zero by the compact support of $\mu_{u^-}(\chi)$. For instance,

$$\text{(102)} \qquad \frac{1}{2\pi}\int_{-\bar{\zeta}}^{\bar{\zeta}} \mu_{u^-}(-\chi)\hat{\phi}_W(\chi)d\chi = \frac{1}{2\pi}\int_{-\bar{\zeta}}^{\bar{\zeta}} \left(\frac{d^{-s}\mu_{u^-}(-\chi)}{d\chi^{-s}}\right)\left(\frac{d^s}{d\chi^s}\hat{\phi}_W(\chi)\right)d\chi,$$

where $d^{-s}\mu_{u^-}(\chi)/d\chi^{-s} = \int_0^{\chi}\dots\int_0^{\zeta_3}\int_0^{\zeta_2} \mu_{u^-}(\zeta_1)d\zeta_1 d\zeta_2 \dots d\zeta_s$ for s as in Definition 2. Similar results hold with $\hat{\phi}_W(\chi)$ replaced by $\phi_W(\chi)$ or $\bar{\phi}_W(\chi)$. Since $u \in \mathbb{S} \Rightarrow u^- \in \mathbb{S} \Rightarrow \mu_{u^-} \in \mathcal{D}_{s,\bar{\zeta}}$, iterating the integration by parts s times guarantees that $d^{-s}\mu_{u^-}(-\chi)/d\chi^{-s}$ is absolutely integrable.

We then have

$$\text{(103)} \qquad \begin{aligned} |R_1^-| &= \left|(2\pi)^{-1}\int_{-\bar{\zeta}}^{\bar{\zeta}} \left(\frac{d^{-s}\mu_{u^-}(-\chi)}{d\chi^{-s}}\right)\left(\frac{d^s}{d\chi^s}(\hat{\phi}_W(\chi) - \bar{\phi}_W(\chi))\right)d\chi\right| \\ &\le (2\pi)^{-1}\int_{-\bar{\zeta}}^{\bar{\zeta}} \left|\frac{d^{-s}\mu_{u^-}(-\chi)}{d\chi^{-s}}\right|d\chi \sup_{\chi\in[-\bar{\zeta},\bar{\zeta}]}\left|\frac{d^s}{d\chi^s}(\hat{\phi}_W(\chi) - \bar{\phi}_W(\chi))\right| \end{aligned}$$

where $\int_{-\bar{\zeta}}^{\bar{\zeta}} |d^{-s}\mu_{u^-}(-\chi)/d\chi^{-s}|d\chi$ was just shown to be finite and it remains to be shown that $\sup_{\chi\in[-\bar{\zeta},\bar{\zeta}]} |d^s(\hat{\phi}_W(\chi) - \bar{\phi}_W(\chi))/d\chi^s| = o_p(n^{-1/2})$. We need to calculate the derivatives $d^s\hat{\phi}_W(\chi)/d\chi^s$ up to the appropriate order s:

$$\text{(104)} \qquad \begin{aligned} \frac{d^s\hat{\phi}_W(\chi)}{d\chi^s} &= \frac{d^s}{d\chi^s}\left(\frac{\hat{m}_W(\chi)}{\hat{m}_1(\chi)}\exp\left(\int_0^{\chi}\frac{i\hat{m}_x(\zeta)}{\hat{m}_1(\zeta)}d\zeta\right)\right) \\ &= \frac{d^{s-1}}{d\chi^{s-1}}\left(\hat{\phi}_1(\chi)\left(\frac{i\hat{m}_{zW}(\chi)}{\hat{m}_1(\chi)} - \frac{\hat{m}_W(\chi)i\hat{m}_z(\chi)}{(\hat{m}_1(\chi))^2} + \frac{\hat{m}_W(\chi)}{\hat{m}_1(\chi)}\frac{i\hat{m}_x(\chi)}{\hat{m}_1(\chi)}\right)\right), \end{aligned}$$

where we have used the fact that $\hat{\phi}_1(\chi) = \exp(\int_0^{\chi}(i\hat{m}_x(\zeta)/\hat{m}_1(\zeta))d\zeta)$. Iterating this process reveals that $d^s\hat{\phi}_W(\chi)/d\chi^s$ is a finite linear combination of terms of the form

$$\text{(105)} \qquad \exp\left(\int_0^{\chi}\frac{i\hat{m}_x(\zeta)}{\hat{m}_1(\zeta)}d\zeta\right)\prod_{t=1}^{\bar{t}}\frac{\hat{m}_{z^{s_t}a_t}(\chi)}{\hat{m}_1(\chi)}$$

where $\bar{t} \le s$ and for $t = 1, \dots, \bar{t}$, each a_t is one of $1, x, w_0, \dots, w_L$ and each $s_t \in \mathbb{N}^+$ is such that $s_t \le s$. The $d^s(\hat{\phi}_W(\chi) - \bar{\phi}_W(\chi))/d\chi^s$ term can then be calculated by substituting expansions (89) and (92) with $a = z^{s_t}a_t$ into equation (105) and by removing the terms that are linear in one of the $\delta\hat{m}_{z^{s_t}a_t}(\zeta)$. Without giving the explicit form of the resulting expansion, it can easily be seen that it has the following properties, which are sufficient to show the negligibility of R_1^- (a simplification made possible by the fact that the R_1^- depends on the behavior of various random functions only over a fixed finite interval, instead of the expanding interval used in the case of R_1^+): Each one of the finite number of terms in the expansion of $d^s(\hat{\phi}_W(\chi) - \bar{\phi}_W(\chi))/d\chi^s$ contains (i) a single $\phi_1(\chi)$, (ii) products of a finite number of moments of the form $m_{z^{s_t}a_t}(\chi)$, (iii) a division by finite powers of $m_1(\chi)$, (iv) a product of at least two (but finitely many) moments of the form $\delta\hat{m}_{z^{s_t}a_t}(\chi)$, (v) terms of the form $(1 + \delta\hat{m}_1(\zeta)/m_1(\zeta))^{-1}$, and (vi) mean values that are smaller in magnitude than the aforementioned quantities. We then note that (i) $|\phi_1(\chi)| \le 1$, (ii) $|m_{z^{s_t}a_t}(\chi)| \le E[|z|^{s_t}|a_t|] \le (E[|z|^{2s_t}]E[|a_t|^2])^{1/2} \le ((E[|z|^{2s}])^{s_t/s}E[|a_t|^2])^{1/2} < \infty$ by the Cauchy–Schwartz and Jensen's inequalities, Assumption 4 and $E[|z|^{2s}] < \infty$ from Definition 2, (iii) $|m_1(\chi)|^{-1} \le A\lambda(\bar{\zeta}) < \infty$ for some $A > 0$ by Lemma 2 and Assumption 5, (iv) by Lemma 6 with $A = \bar{\zeta}$ and $\kappa = 0$, $\sup_{\chi\in[-\bar{\zeta},\bar{\zeta}]} |\delta\hat{m}_{z^{s_t}a_t}(\chi)| = o_p(n^{-1/2+\varepsilon})$ for any $\varepsilon > 0$, if $E[|a_t|^2|z|^{2s_t}]$ and $E[|a_t||z|^{s_t+1}]$ exist, which is implied by Assumption 4 and $E[|z|^{2s}]$, $E[|a_t|^2|z|^{2s}] < \infty$ from Definition 2, by Hölder's inequality,[17] (v) $(1 + \delta\hat{m}_1(\zeta)/m_1(\zeta))^{-1} = O_p(1)$ since $|\delta\hat{m}_1(\chi)| \le \hat{M}_n = o_p(1)$ and $|m_1(\chi)|^{-1} \le A\lambda(\bar{\zeta}) < \infty$, for some $A > 0$. Collecting these orders of magnitude, we obtain

$$\sup_{\chi\in[-\bar{\zeta},\bar{\zeta}]} \left|d^s\big(\hat{\phi}_W(\chi) - \bar{\phi}_W(\chi)\big)/d\chi^s\right| = O(1)\big(o_p(n^{-1/2+\varepsilon})\big)^2 O_p(1)$$

$$= o_p(n^{-1+2\varepsilon}) = o_p(n^{-1/2}),$$

thus implying that $R_1^- = o_p(n^{-1/2})$, as desired.

Now consider the Ψ^{W,u^-} term, which can be rewritten as

$$(2\pi)^{-1}\int_{-\bar{\zeta}}^{\bar{\zeta}} \big(d^{-s}\mu_{u^-}(-\chi)/d\chi^{-s}\big)\big(d^s\big(\bar{\phi}_W(\chi) - \phi_W(\chi)\big)/d\chi^s\big)d\chi.$$

The term $d^s(\bar{\phi}_W(\chi) - \phi_W(\chi))/d\chi^s$ can be calculated by substituting expansions (89) and (92) with $a = z^{s_t}a_t$ into equation (105) and by *keeping* only the terms that are linear in one of the $\delta\hat{m}_{z^{s_t}a_t}(\zeta)$. Once again, without explicitly giving the expression of such an expansion, it can easily be seen that each term has the general form $\int U_{z^{s_t}a_t}(\zeta)\delta\hat{m}_{z^{s_t}a_t}(\zeta)d\zeta$ where $U_{z^{s_t}a_t}(\zeta)$ contains (i) a single $\phi_1(\chi)$, (ii) products of a finite number of moments of the form $m_{z^{s_t}a_t}(\zeta)$, (iii) a division by finite powers of $m_1(\zeta)$, (iv) a single $d^{-s}\mu_{u^-}(-\chi)/d\chi^{-s}$. The quantities (i), (ii), and (iii) are bounded by the same argument as for the R_1^- term, while (iv) $d^{-s}\mu_{u^-}(-\chi)/d\chi^{-s}$ is absolutely integrable. It follows that $U_{z^{s_t}a_t}(\zeta)$ is absolutely integrable. After noting that all the $E[|a_t|^2|z|^{2s_t}]$ exist (as shown earlier) and that integration by parts preserves the parity of the integrand, implying that Ψ^{W,u^-} is real-valued, we can invoke Lemma 4 with $a = z^{s_t}a_t$ to show that $\int U_{z^{s_t}a_t}(\zeta)\delta\hat{m}_{z^{s_t}a_t}(\zeta)d\zeta$ is asymptotically normal and root n consistent.

Having shown that Ψ^{W,u^+} and Ψ^{W,u^-} are asymptotically normal and root n consistent, we calculate the asymptotic variance of $(\Psi^{W,u^+} + \Psi^{W,u^-}) \equiv \Psi^{W,u}$ or, more generally, the asymptotic covariance between Ψ^{W_1,u_1} and Ψ^{W_2,u_2} for $W_1, W_2 = 1, w_0, \ldots, w_L$ and $u_1, u_2 \in \mathbb{S}$. Following the same steps as in equation (85) while replacing u^+ by u, we find that $\Psi^{W,u} = \sum_{a=1,x,W}\int U_a^{W,u}(\zeta)\delta\hat{m}_a(\zeta)d\zeta$, for $U_a^{W,u}(\zeta)$ as in the statement of the theorem, where it is understood that integrals involving a singular $\mu_u(\chi)$ are to be evaluated by parts, so that all integrands can be made absolutely integrable, thus enabling the required permutation among integration and expectation operations. We can then write

$$E\big[\Psi^{W_1,u_1}\Psi^{W_2,u_2}\big] = \sum_{a=1,x,W_1}\sum_{b=1,x,W_2}\int U_a^{W_1,u_1}(\zeta)E\big[\delta\hat{m}_a(\zeta)\delta\hat{m}_b^\dagger(\xi)\big]\big(U_b^{W_2,u_2}(\xi)\big)^\dagger d\zeta d\xi. \tag{106}$$

[17] $E[|a_t||z|^{s_t+1}] \le (E[|a_t|^2|z|^{2s_t}]E[|z|^2])^{1/2} \le (E[|a_t|^2|z|^{2s_t}])^{1/2}(E[|z|^{2s}])^{1/(2s)}$ and $E[|a_t|^2|z|^{2s_t}] = E[(|a_t|^{2-2s_t/s})(|a_t|^{2s_t/s}|z|^{2s_t})] \le (E[|a_t|^2])^{(1-s_t/s)}(E[|a_t|^2|z|^{2s}])^{s_t/s}$.

Proceeding as for the calculations of $V_{ab}(\zeta, \xi)$ in Lemma 4 then provides the asymptotic variance given in the statement of the theorem, while taking the convention that all the integrals involving distributions are to be evaluated via integration by parts. *Q.E.D.*

PROOF OF THEOREM 4: By Theorem 2, Assumptions 1 through 6 imply that the elements of $\hat{C}$ that involve a function of x^* are asymptotically normal and root n consistent, while Assumption 7 lets us reach the same conclusion for the elements of $\hat{C}$ that do not involve a function of x^*. Assumptions 1 through 6 also imply that the elements of $\hat{D}$ that involve a function of x^* are asymptotically normal and root n consistent while Assumption 7 implies consistency of the remaining elements of $\hat{D}$. The first order change in β when D and C change by an amount δD and δC, respectively, is[18] $\delta\beta = -D^{-1}(\delta D)D^{-1}C + D^{-1}\delta C$, which can be written, for $k = 1, \dots, M+L$, as

$$\begin{aligned}(107)\qquad \delta\beta_k &= -\sum_{j=1}^{(M+L)}\sum_{l=1}^{(M+L)}(D^{-1})_{kj}(\delta D)_{jl}(D^{-1}C)_l + \sum_{j=1}^{(M+L)}(D^{-1})_{kj}\delta C_j \\ &= -\sum_{j=1}^{(M+L)}\sum_{l=1}^{(M+L)}(D^{-1}C)_l(D^{-1})_{kj}(\delta D)_{jl} + \sum_{j=1}^{(M+L)}(D^{-1})_{kj}\delta C_j,\end{aligned}$$

or $\delta\beta = P\,\delta p$ where $P = [-(C'D^{-1}) \otimes D^{-1}, D^{-1}]$ and $\delta p = [\text{vec}(\delta D)', \delta C']'$. This implies that $\hat{\beta}$ is a differentiable function of $\hat{D}$ and $\hat{C}$ at $\hat{D} = D$ and $\hat{C} = C$. By the delta method, these results imply asymptotic normality and root n consistency of $\hat{\beta}$ and the asymptotic variance is given by $\text{Var}(\hat{\beta}) = P\,\text{Var}(\delta p)P'$ where $\text{Var}(\delta p)$ is the $((M+L)^2 + (M+L)) \times ((M+L)^2 + (M+L))$ matrix of all covariances between any two elements of the vector δp, as given by equation (33). *Q.E.D.*

PROOF OF LEMMA 1: Corollary 2.2 in Newey (1991) (with his $h(x) \equiv x$) provides sufficient conditions for uniform convergence in probability on a compact set, which we verify in turn. (i) $(\mathcal{C}, d)$ is compact by assumption. (ii) The fact that $|\tilde{E}[u(x^*, w, \beta)] - E[u(x^*, w, \beta)]| \xrightarrow{p} 0$ pointwise follows from the Assumption that $u \in \mathbb{S}'_{\mathcal{C}}$ and Theorem 6, which shows (root n) consistency of $\tilde{E}[u(x^*, w, \beta)]$ for $\beta \in \mathcal{C}$, (iii) $|\tilde{E}[u(x^*, w, \gamma)] - \tilde{E}[u(x^*, w, \beta)]| \le \hat{u}d(\gamma, \beta)$ with $\hat{u} = O_p(1)$ since equation (62) implies that

$$(108)\qquad -\bar{u}(x^*, w)d(\gamma, \beta) \le u(x^*, w, \gamma) - u(x^*, w, \beta) \le \bar{u}(x^*, w)d(\gamma, \beta),$$

inequalities which are left unchanged by taking estimated expectations $\tilde{E}[\cdot]$ of each term. Since $\bar{u} \in \mathbb{S}'_{\emptyset}$, by Theorem 6, $\hat{u} = \tilde{E}[\bar{u}(x^*, w)] = E[\bar{u}(x^*, w)] + O_p(n^{-1/2}) = O_p(1)$ and the desired conclusion follows. *Q.E.D.*

LEMMA 7: *Under Assumptions* 8 *through* 17, $\hat{\beta} \xrightarrow{p} \beta^*$ *where* $\hat{\beta}$ *is given by equation* (39).

[18]Note that

$$\begin{aligned}(D+\delta D)^{-1}(C+\delta C) - D^{-1}C &= (I + D^{-1}\delta D)^{-1}D^{-1}(C+\delta C) - D^{-1}C \\ &= (I - D^{-1}\delta D)D^{-1}(C+\delta C) - D^{-1}C + o(\|\delta D\|) \\ &= D^{-1}C - D^{-1}(\delta D)D^{-1}C + D^{-1}\delta C - D^{-1}C \\ &\quad + o(\|\delta D\|).\end{aligned}$$

PROOF: Theorem 2.1 in Newey and McFadden (1994) states four sufficient conditions for consistency, which we can easily verify. By Assumption 14, (i) the objective function is uniquely maximized at $\beta = \beta^*$ and (ii) β^* lies in the interior of a compact set $\mathcal{B}$. (iii) Both $H(\cdot)$ and $r(\beta)$ are continuous by Assumptions 15 and 16, implying that $H(r(\beta))$ is continuous in β. (iv) Finally, we need to show that $\sup_{\beta\in\mathcal{B}} |H(\tilde{E}[R(x^*, w, \beta)]) - H(E[R(x^*, w, \beta)])| \xrightarrow{p} 0$. By Assumption 17 and Lemma 1, $\sup_{\beta\in\mathcal{B}} |\tilde{E}[R(x^*, w, \beta)] - E[R(x^*, w, \beta)]| \xrightarrow{p} 0$. The uniform convergence of $\tilde{E}[R(x^*, w, \beta)]$ implies uniform convergence of $H(\tilde{E}[R(x^*, w, \beta)])$ since we can show that $H(\cdot)$ is uniformly continuous over $\mathcal{R}$, the image of $\mathcal{B}$ by $r(\beta)$. The derivative of $r(\beta)$ is continuous and therefore reaches its maximum in the compact set $\mathcal{B}$. It follows that $r(\beta)$ is uniformly continuous on $\mathcal{B}$ and that $\mathcal{R}$ must be compact. Since the derivative of $H(r)$ is also continuous over the compact set $\mathcal{R}$, by the same reasoning, $H(r)$ is also uniformly continuous over $\mathcal{R}$. *Q.E.D.*

PROOF OF THEOREM 7: For conciseness, some of the conditions of the theorem are stronger than necessary. In particular, we use the root n consistency provided by Theorem 6 whenever we need consistency or boundedness in probability. Denote all quantities estimated from Definition 7 by "$\sim$". The first order condition to find $\hat{\beta}$ is $q(\tilde{g}(\hat{\beta})) = 0$, where the elements of $q(\tilde{g}(\hat{\beta}))$ are given by

$$q_l(\tilde{g}(\hat{\beta})) \equiv \sum_{j=1}^{J} \frac{\partial H(\tilde{r}(\hat{\beta}))}{\partial r_j} \frac{\partial \tilde{r}_j(\hat{\beta})}{\partial \beta_l} \quad \text{for } l = 1, \ldots, B. \tag{109}$$

By a Taylor expansion around β^*, we have $q(\tilde{g}(\beta^*)) + Q(\tilde{g}(\bar{\beta}))\tilde{G}(\bar{\beta})(\hat{\beta} - \beta^*) = 0$ where $\bar{\beta}$ lies on the segment joining β^* and $\hat{\beta}$. Expanding $q(\tilde{g}(\beta^*))$ around $\tilde{g}(\beta^*) = g(\beta^*)$, we obtain: $q(g(\beta^*)) + Q(\bar{g})(\tilde{g}(\beta^*) - g(\beta^*)) + Q(\tilde{g}(\bar{\beta}))\tilde{G}(\bar{\beta})(\hat{\beta} - \beta^*) = 0$, where $\bar{g}$ lies on the segment joining $\tilde{g}(\beta^*)$ and $g(\beta^*)$. By definition of β^*, $q(g(\beta^*)) = 0$ and rearranging yields: $(\hat{\beta} - \beta^*) = -(Q(\tilde{g}(\bar{\beta}))\tilde{G}(\bar{\beta}))^{-1}Q(\bar{g})(\tilde{g}(\beta^*) - g(\beta^*))$. Since, by Lemma 7, $\hat{\beta} \xrightarrow{p} \beta^*$ and thus $\bar{\beta} \xrightarrow{p} \beta^*$, we have that $\hat{\beta}, \bar{\beta} \in \mathcal{N}$ with probability approaching 1. Now, by Assumption 19 and Lemma 1, both $\tilde{g}(\beta) \xrightarrow{p} g(\beta)$ and $\tilde{G}(\beta) \xrightarrow{p} G(\beta)$ uniformly for $\beta \in \mathcal{N}$, and thus $\bar{g} \xrightarrow{p} g(\beta^*)$. These facts, along with the continuity of $Q(g)$, $G(\beta)$, and $g(\beta)$, establish that $Q(\tilde{g}(\bar{\beta}))\tilde{G}(\bar{\beta}) \xrightarrow{p} Q(g(\beta^*))G(\beta^*) \equiv QG$ and that $Q(\bar{g}) \xrightarrow{p} Q(g(\beta^*)) \equiv Q$. Under Assumption 18, by continuity of the inverse $(Q(\tilde{g}(\bar{\beta}))\tilde{G}(\bar{\beta}))^{-1} \xrightarrow{p} (QG)^{-1}$. By Theorem 6 and the Slutzky Theorem, $n^{1/2}(\hat{\beta} - \beta^*) \xrightarrow{d} N(0, (QG)^{-1}Q\Gamma Q'(G'Q')^{-1})$, where element Γ_{kl} of the matrix Γ is given by equation (61) for $\gamma = \beta = \beta^*$ and $u_1(x^*, w, \beta^*) = g_k(x^*, w, \beta^*)$ and $u_2(x^*, w, \beta^*) = g_l(x^*, w, \beta^*)$ for $k, l = 1, \ldots, J(B+1)$. *Q.E.D.*

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